NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

CHAPTER 1

# Counting

Discrete Mathematics 1 | First year

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# Roadmap of the chapter

1. Introduction: what is discrete mathematics?
2. Fundamental principles: product or sum?
3. Distinguishability, ordering and repetition
4. Arrangements and permutations
5. Combinations, binomial theorem, Pascal's triangle
6. Multinomial coefficients and theorem
7. Combinations with repetition

**Convention for this course**

Statements and examples are on the slides; **proofs are done on the board**

✎ **Proof on the board**

01

# Introduction

What is discrete mathematics, and why do we count?

# What is Discrete Mathematics?

**Definition**

**Discrete Mathematics** is a field of mathematics that deals with distinct and countable objects, rather than quantities that change continuously.

- It studies structures that are fundamentally discrete: numbers, logical statements, algebraic structures, ...
- It forms the backbone of **computer science**, **cryptography**, and many branches of **pure mathematics**.

# Key concepts in Discrete Mathematics

- **Sets:** collections of distinct objects.
- **Logic:** principles of reasoning and valid arguments.
- **Algebraic structures:** groups, rings, fields.
- **Number theory:** properties of the integers.
- **Graph theory:** vertices joined by edges; networks, relations, algorithms.

**Combinatorics**

The art of **counting, organizing and selecting** objects. We focus on enumerative combinatorics. Key questions:

- Is the configuration possible?
- In how many ways can it be made?
- How do we find such a configuration?

# Some questions in combinatorics

- How many distinct ways can the letters of the word "NHSM" be arranged?
- 12 students, 3 different prizes (gold, silver, bronze): in how many ways can the prizes be distributed?
- How many ways can you arrange 6 balls into 4 boxes?
- How many ways can you assign 5 distinct colors to 5 identical chairs in a row?
- How many passwords of length $n$ are possible using 26 letters?
- 7 tasks, 3 employees: in how many ways can the tasks be assigned if each employee gets at least one task?

# Listing versus counting

**Remark**

When doing enumeration, it is very important to understand the distinction between listing all the configurations and determining their number.



# Why study combinatorics?

- **Computer science:** efficient algorithms, data structures, cryptographic systems, programming, network theory.
- **Engineering:** optimizing system designs, network configurations, resource allocations.
- **Mathematics:** number theory, algebraic structures, discrete optimization.
- **Statistics:** the foundation of probability theory and statistical models.
- **Physics:** quantum mechanics and statistical physics, where discrete models are essential.

*Powerful techniques for logical reasoning, problem-solving and analyzing complex systems.*

# A first exercise

✎ **Exercise**

How many different 2-letter words, including nonsense words, can be produced using the letters $\{H, I, M\}$ (letters may be repeated)?

# A first exercise: solution

**Solution**

The set of all possible 2-letter words, including nonsense words, is

$$\{HI, HM, IM, IH, MH, MI, II, HH, MM\};$$

there are 9 such words.

# Let us change the question a little

**✎ Exercise**

How many different 26-letter words, including nonsense words, can be produced using the letters of the alphabet (letters may be repeated)?

# Let us change the question a little: solution

**Solution**

There are $26^{26}$ such words. Even using a computer that prints 1 billion words per second, it will take about

$$1.95 \times 10^{18} \text{ centuries}$$

to print all the possible words.

02

# Fundamental Principles of Counting

Product or sum?

# Cardinality of a set

**Definition**

The **cardinality** of a finite set $S$ is the number of its elements. It is denoted by $|S|$.

**Examples**

$$S = \{a, b, c, d\} \implies |S| = 4, \qquad T = \{1, 2, 3, 4, 5, 6, 7\} \implies |T| = 7.$$

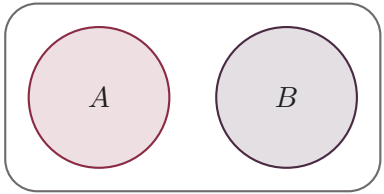
# Addition principle

**Theorem (Addition Principle)**

If $A$ and $B$ are two **disjoint** sets ($A \cap B = \emptyset$), then

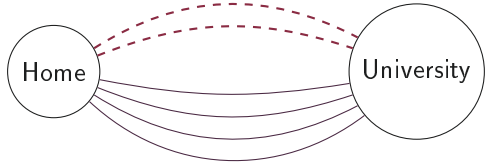
$$|A \cup B| = |A| + |B|.$$

✎ **Proof on the board**



# Addition principle: examples

1. 2 different train routes and 4 different bus routes from home to the university.
   With $A$ = train routes, $B$ = bus routes: $|A| + |B| = 6$ possible routes.




2. A class has 8 boys and 5 girls; we choose one representative:
   $|\text{Boys}| + |\text{Girls}| = 13$ possible choices.

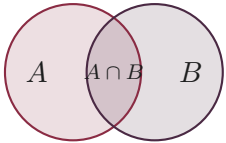
# Disjointness is essential

**Remark**

If $A \cap B \neq \emptyset$, the elements of $A \cap B$ are counted twice, and

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

This is the starting point of the **Inclusion–Exclusion Principle**.



# Generalised addition principle

**Corollary (Generalised Addition Principle)**

If $A_1, A_2, \ldots, A_n$ are $n$ pairwise disjoint sets ($A_i \cap A_j = \emptyset$ for all $i \neq j$), then

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} |A_i|.$$

✎ Proof by induction on the board

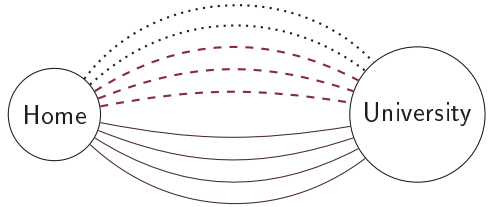
# Generalised addition principle: example

Three modes of transportation:

- 3 train routes ($A_1$), dashed,
- 4 bus routes ($A_2$), solid,
- 2 tram routes ($A_3$), dotted.

The sets are pairwise disjoint (one mode at a time), so

$$|A_1 \cup A_2 \cup A_3| = 3 + 4 + 2 = 9.$$



# Addition principle: alternate form

**Theorem (Addition Principle: Alternate Form)**

If a problem consists of solving **exactly one** of $k$ tasks, where no way of doing one task is also a way of doing another, and each task $i$ $(1 \leq i \leq k)$ can be done in $n_i$ ways, then the total number of solutions is

$$\sum_{i=1}^{k} n_i = n_1 + n_2 + \cdots + n_k.$$

# Multiplication principle

**Definition (Cartesian product)**

$A \times B$ is the set of all ordered pairs $(a, b)$ with $a \in A$ and $b \in B$.

**Theorem (Multiplication Principle)**

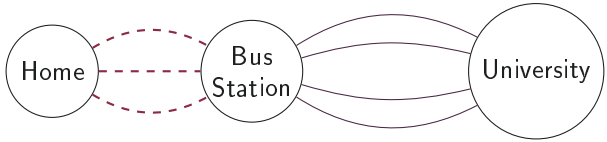
If $A$ and $B$ are two sets, then

$$|A \times B| = |A| \cdot |B|.$$

✎ Proof on the board

# Multiplication principle: examples

1. Choose one of 3 train routes from home to the bus station, **then** one of 4 bus routes from the bus station to the university: $3 \times 4 = 12$ routes.




2. 8 boys and 5 girls; a committee of one boy **and** one girl: $8 \times 5 = 40$ ways.

# Generalised multiplication principle

**Corollary (Generalised Multiplication Principle)**

If $A_1, A_2, \ldots, A_n$ are $n$ sets, then

$$|A_1 \times A_2 \times \cdots \times A_n| = |A_1| \cdot |A_2| \cdots |A_n|.$$

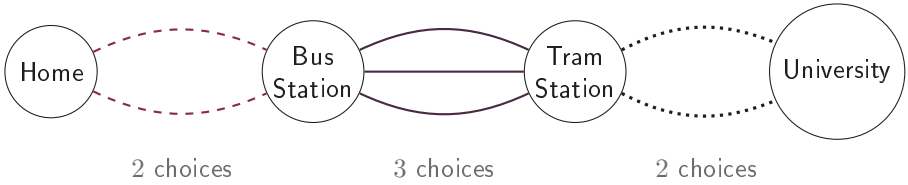
✎ Proof by induction on the board

**Example**

Home → bus station: one of 2 train routes; bus station → tram station: one of 3 bus routes; tram station → university: one of 2 tram routes.

$$|A_1 \times A_2 \times A_3| = 2 \times 3 \times 2 = 12 \text{ ways.}$$

# A journey in three stages




Total: $2 \times 3 \times 2 = 12$ different ways.

# Multiplication principle: alternate form

**Theorem (Multiplication Principle: Alternate Form)**

If a problem consists of solving $k$ tasks **sequentially**, and each task $i$ $(1 \leq i \leq k)$ can be performed in $n_i$ ways, then the total number of solutions is

$$\prod_{i=1}^{k} n_i = n_1 \times n_2 \times \cdots \times n_k.$$

**Remark**

The **number** $n_i$ of ways to perform task $i$ must not depend on the choices made in the previous tasks (the choices themselves may change, but not their number).

# Product or sum?

**Rule of thumb**

"**or**" (one case among disjoint cases) $\Longrightarrow$ add; "**and then**" (successive steps) $\Longrightarrow$ multiply.

**Example**

How many passwords of length 1, 2 or 3 can be formed with 26 letters?

A password has length 1 **or** 2 **or** 3 (addition), and a password of length $i$ is $i$ successive choices (multiplication):

$$26 + 26^2 + 26^3 = 18\,278.$$

03

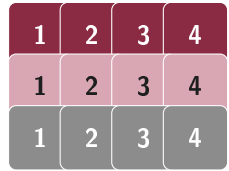
# Distinguishability, Ordering & Repetition

Three questions to ask before counting anything

# Distinguishability

**Definition (Distinguishability)**

Two elements of $S$ are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

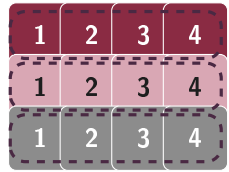


# Distinguishability

**Definition (Distinguishability)**

Two elements of $S$ are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.



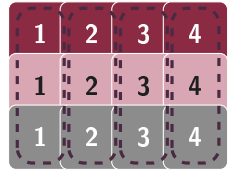
3 groups (colors)

# Distinguishability

**Definition (Distinguishability)**

Two elements of $S$ are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.
- **By number:** 4 distinguishable groups, each with 3 indistinguishable squares.



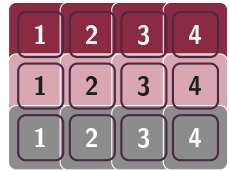
4 groups (numbers)

# Distinguishability

**Definition (Distinguishability)**

Two elements of $S$ are **indistinguishable** if they are identical with respect to the property in question; otherwise they are **distinguishable**.

- **By color:** 3 distinguishable groups, each with 4 indistinguishable squares.
- **By number:** 4 distinguishable groups, each with 3 indistinguishable squares.
- **By color and number:** 12 pairwise distinguishable squares.



12 different squares

# Ordering

**Definition (Ordering)**

If the selected elements are treated equally and play the same role, the selection is **unordered**. If they can occupy different roles (their positions matter), the selection is **ordered** (a list).

**Example**

The 100-meter final at the Olympics, with 8 participants:

- select 3 runners for a doping test: unordered;
- select 3 runners for gold, silver and bronze medals: ordered, since the positions matter.

*If the items are truly indistinguishable, then "order" becomes meaningless!*

# Ordered or unordered? A picture

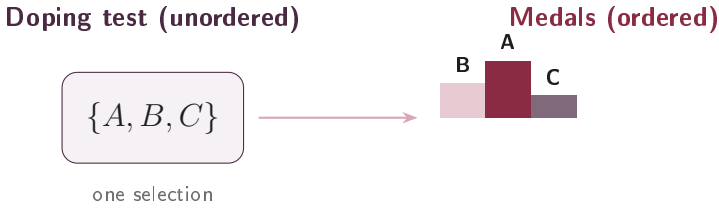
**Doping test (unordered)**

$\{A, B, C\}$

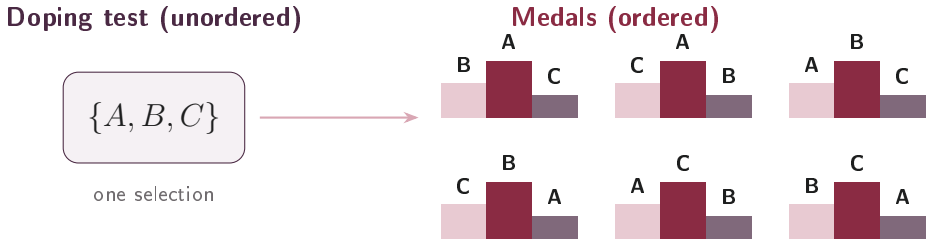
one selection

**Medals (ordered)**

# Ordered or unordered? A picture



# Ordered or unordered? A picture




$3! = 6$ ordered results for **one** unordered selection

# Repetition

**Definition (Repetition)**

If an element of $S$ can appear more than once in the selection, **repetition is allowed**. If each element can appear only once, repetition is **not allowed**.

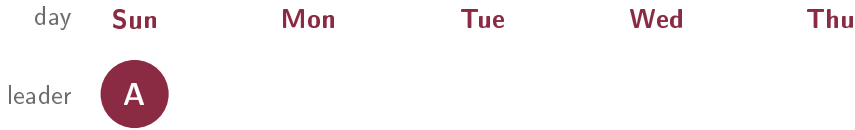
**Example**

We select a leader from a group of students **each day**. Repetition is possible: a student can be chosen more than once.

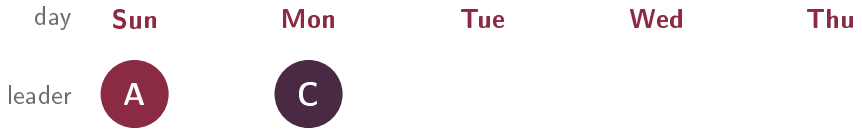
*Does the order matter in this case?*

We now consider selections of $k$ elements under the conditions of **ordering** and **repetition**.

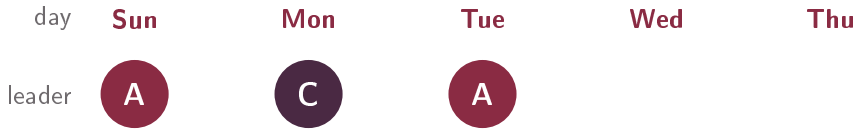
# Repetition: a week of leaders



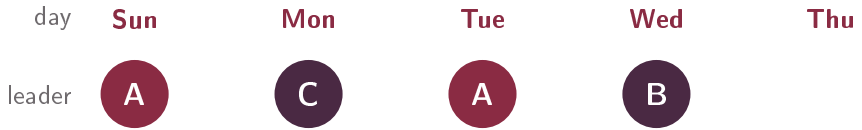
# Repetition: a week of leaders



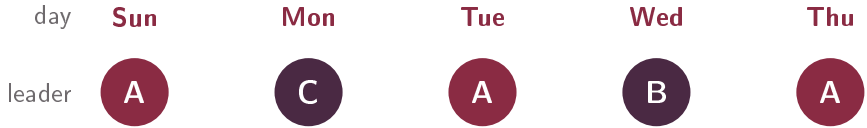
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# Repetition: a week of leaders



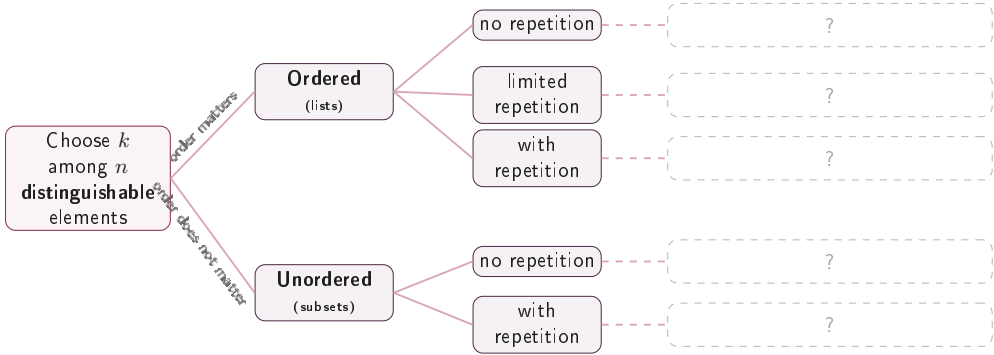
# Repetition: a week of leaders




Student $A$ is chosen **three times**: repetition is allowed.
The days are different positions: the order matters.

# Where are we? The tree of notions

Choose $k$ among $n$ distinguishable elements: **does the order matter? Is repetition allowed?**



# 3.1

## Arrangements & Permutations

Ordered selections, with or without repetition

# Arrangements ($k$-permutations)

**Definition**

Let $S = \{e_1, \ldots, e_n\}$. An **arrangement** (or $k$-**permutation**) of $k$ elements from $S$ is any **ordered** sequence $(a_1, a_2, \ldots, a_k)$ of $k$ elements selected from $S$.

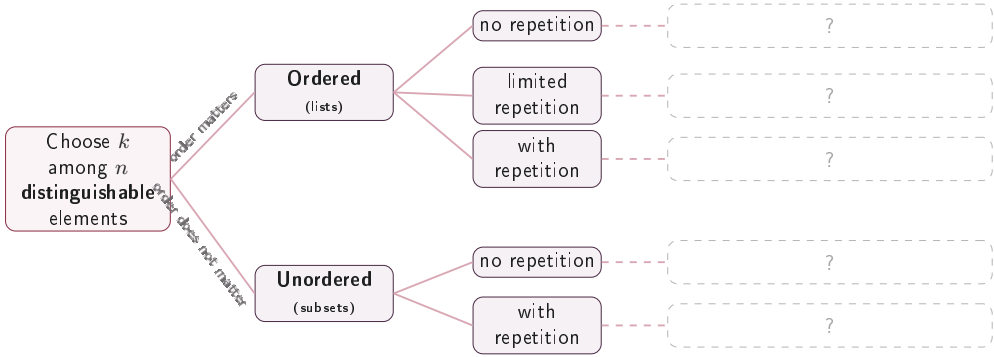
**Examples**

- $S = \{1, 2, \ldots, 9\}$: $123,\ 112,\ 344,\ 789,\ 258$ are arrangements of 3 elements.
- $S = \{A, B, C, D, E, F\}$: $ABC,\ DEF,\ AAB,\ CDC,\ CDA$ are arrangements of 3 elements.

*An arrangement can simply be described as a **listing** of elements.*

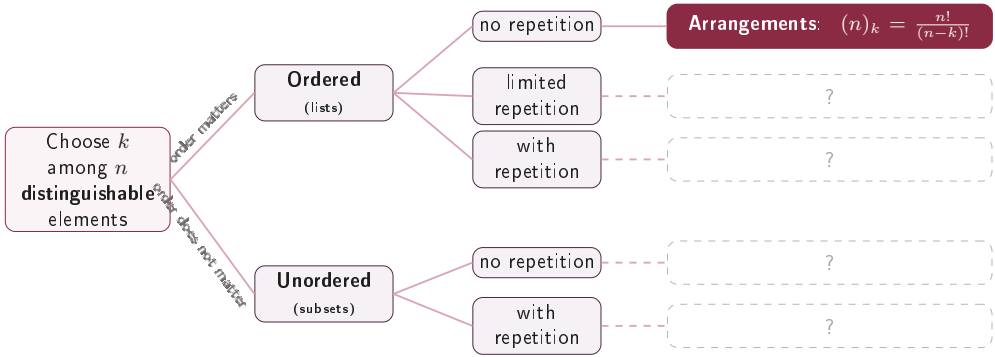
# Where are we? Arrangements

First leaf: **ordered** selections **without repetition**, the arrangements.



# Where are we? Arrangements

First leaf: **ordered** selections **without repetition**, the arrangements.



# $k$-permutations without repetition

**Definition**

A $k$-**permutation without repetition** is an arrangement of $k$ **distinct** elements of $S$: an ordered sequence of $k$ different elements.

**Example**

From the previous examples, $123$, $789$ and $258$ are 3-permutations **without** repetition from the first set, while $AAB$ and $CDC$ are 3-permutations **with** repetition from the second set.

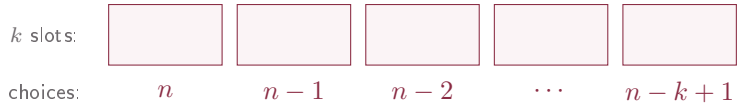
# Counting $k$-permutations without repetition

**Theorem (Number of $k$-permutations without repetition)**

For $0 \leq k \leq n$, the number of $k$-permutations without repetition from a set of $n$ elements, denoted $(n)_k$ (also written $A_n^k$), is

$$(n)_k = n \times (n-1) \times (n-2) \times \cdots \times (n-k+1).$$

✎ Proof on the board



# Exercise

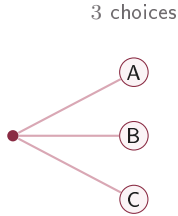
**Exercise**

- Let $S = \{A, B, C\}$. How many 2-permutations without repetition can be formed? List them.
- Let $S = \{A, B, C, D, E, F\}$. How many 4-permutations without repetition can be formed?

# Solution

**Solution**

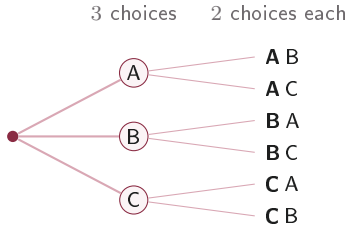
- $(3)_2 = 3 \times 2 = 6$: $AB,\ AC,\ BA,\ BC,\ CA,\ CB$.
- $(6)_4 = 6 \times 5 \times 4 \times 3 = 360$.



# Solution

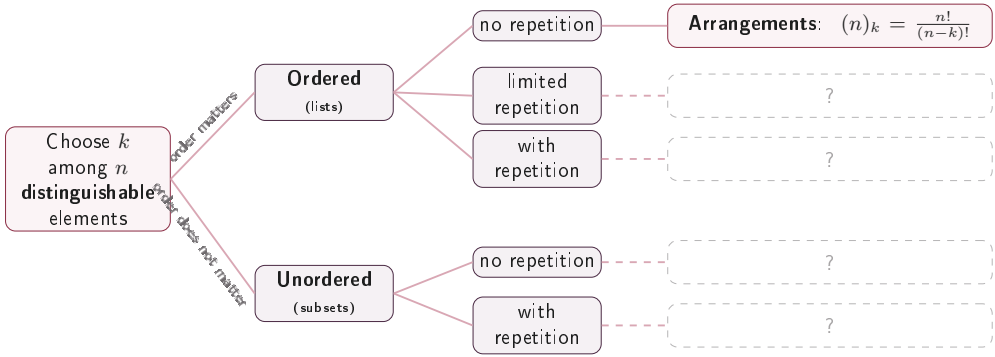
**Solution**

- $(3)_2 = 3 \times 2 = 6$: $AB,\ AC,\ BA,\ BC,\ CA,\ CB$.
- $(6)_4 = 6 \times 5 \times 4 \times 3 = 360$.



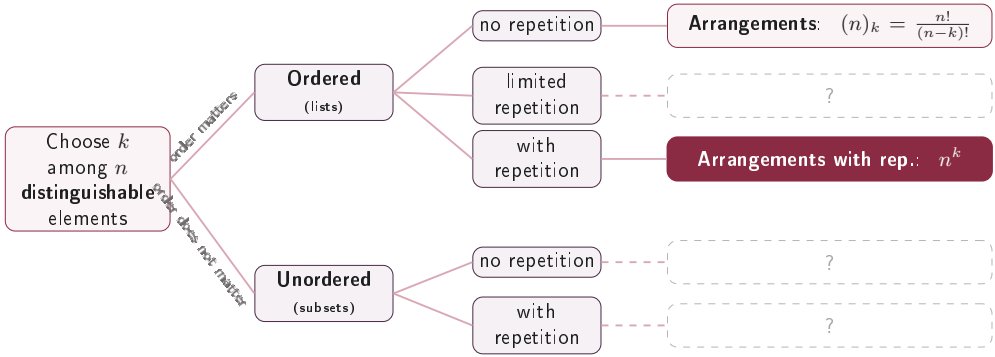
# Where are we? Arrangements with repetition

Second leaf: **ordered** selections **with repetition**.



# Where are we? Arrangements with repetition

Second leaf: **ordered** selections **with repetition**.



# $k$-permutations with repetition

**Definition**

A **$k$-permutation with repetition** is an arrangement of $k$ elements of $S$ where repetition is allowed: each element can appear more than once.

**Example**

$112$ and $344$ are 3-permutations with repetition from the first set, while $AAB$ and $CDC$ are 3-permutations with repetition from the second set.

# Counting $k$-permutations with repetition

**Theorem (Number of $k$-permutations with repetition)**

The number of $k$-permutations with repetition from a set of $n$ elements, denoted $\mathcal{P}(n, k)$, is

$$\mathcal{P}(n, k) = n^k.$$

✎ Proof on the board




The repetition here is allowed and unlimited. Careful: some textbooks (e.g. Rosen) use $P(n, k)$ for $(n)_k$, the count *without* repetition.

# Exercise

**✎ Exercise**

Let $S = \{1, 2, 3\}$. How many 2-permutations with repetition can be formed? List them.

# Solution

**Solution**

Here $n = 3$ and $k = 2$, so $\mathcal{P}(3,2) = 3^2 = 9$:

$$11,\ 12,\ 13,\ 21,\ 22,\ 23,\ 31,\ 32,\ 33.$$

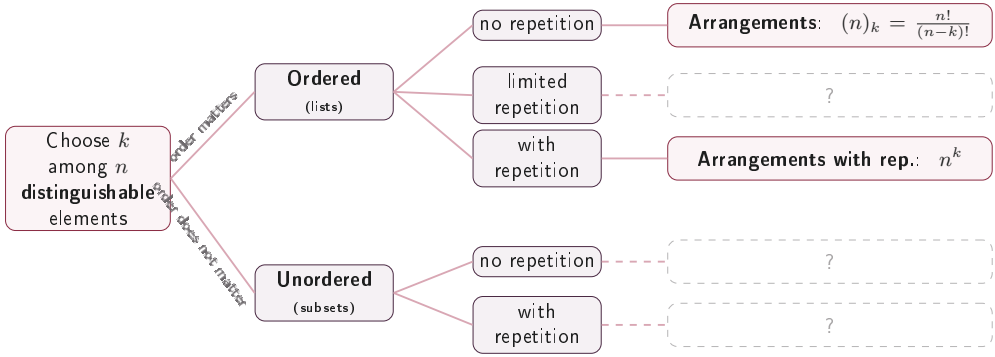
This is exactly the $\{H, I, M\}$ exercise of the introduction.

| 1st \ 2nd | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 11 | 12 | 13 |
| 2 | 21 | 22 | 23 |
| 3 | 31 | 32 | 33 |

$3 \times 3 = 9$

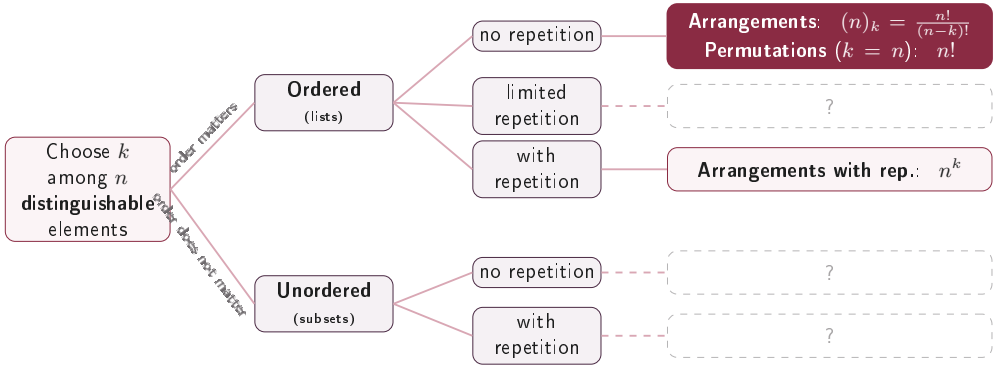
# Where are we? Permutations

Special case of the first leaf: arrange **all** the $n$ elements ($k = n$).



# Where are we? Permutations

Special case of the first leaf: arrange **all** the $n$ elements ($k = n$).



# Permutations (without repetition)

For a non-negative integer $n$, the **factorial** of $n$ is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1, \qquad 0! = 1$$

($0!$ is the empty product; it keeps the rule $n! = n\,(n-1)!$ valid for $n = 1$.)

**Definition**

A **permutation** of $S = \{e_1, \ldots, e_n\}$ is an arrangement of **all** the elements of $S$ without repetition (an $n$-permutation without repetition).

**Example**

$S = \{A, B, C\}$: $ABC,\ ACB,\ BAC,\ BCA,\ CAB,\ CBA,$ so 6 permutations.

# Number of permutations

**Theorem (Number of Permutations)**

The number of permutations of a set with $n$ elements, denoted $P_n$, is

$$P_n = n!.$$

✎ Proof on the board

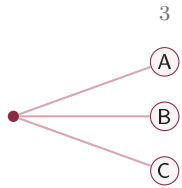
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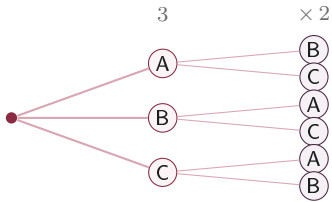
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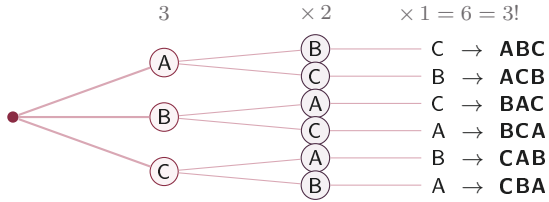
# Number of permutations

**Theorem (Number of Permutations)**

The number of permutations of a set with $n$ elements, denoted $P_n$, is

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✎ Proof on the board



# Exercise

**✎ Exercise**

How many nonsense words of length 26 can be formed using all the letters of the English alphabet without repetition?

# Solution

**Solution**

This is a permutation of the 26 letters: $P_{26} = 26! \approx 4.03 \times 10^{26}$.
Compare with the $26^{26} \approx 6.16 \times 10^{36}$ words of the introduction, where repetition was allowed.

# Relations between the counts

**Proposition**

- For all $n \geq 0$: $P_n = (n)_n = n!$.
- For $0 \leq k \leq n$: $(n)_k = \dfrac{n!}{(n-k)!}$.
- $P_n = (n)_k \, P_{n-k}$.

✎ **Exercise**

Prove the proposition.

# Solution: Relations between the counts

**Solution**

- A permutation of $S$ is an $n$-permutation without repetition, so $P_n = (n)_n = n(n-1)\cdots 1 = n!$.
- For $0 \le k \le n$, split the product $n!$ after its first $k$ factors:

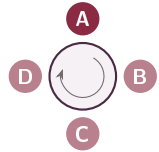
$$n! = \underbrace{n(n-1)\cdots(n-k+1)}_{(n)_k} \cdot \underbrace{(n-k)(n-k-1)\cdots 1}_{(n-k)!}, \qquad \text{so} \qquad (n)_k = \frac{n!}{(n-k)!}$$

  (for $k = n$ the second product is empty and equals $0! = 1$).
- This is the same identity: $P_n = n! = (n)_k\,(n-k)! = (n)_k\,P_{n-k}$.
  Combinatorially: to arrange the $n$ elements, fill the first $k$ positions ($(n)_k$ ways), then arrange the $n-k$ remaining elements in the last $n-k$ positions ($P_{n-k}$ ways), and use the multiplication principle.

# Circular permutations

**Definition (Circular permutation)**

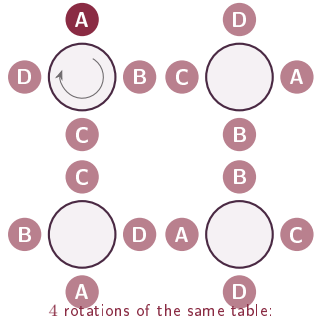
A **circular permutation** of $n$ distinct elements is a way to place them **around a circle** (for example, people around a round table).
Two placements are the **same** if one is a **rotation** of the other: everybody moves by the same number of seats, so everybody keeps the **same neighbours**.



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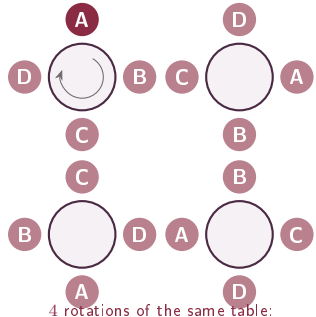

4 rotations of the same table:
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A **circular permutation** of $n$ distinct elements is a way to place them **around a circle** (for example, people around a round table).
Two placements are the **same** if one is a **rotation** of the other: everybody moves by the same number of seats, so everybody keeps the **same neighbours**.

In a **line**, $ABCD$, $DABC$, $CDAB$, $BCDA$ are $4$ **different** permutations. Around the table, they are the **same** circular permutation: we do not count the rotations.



4 rotations of the same table:
**one** circular permutation

# Number of circular permutations

**Theorem (Number of circular permutations)**

For $n \geq 1$, the number of circular permutations of $n$ distinct elements is $\frac{n!}{n} = (n-1)!$.

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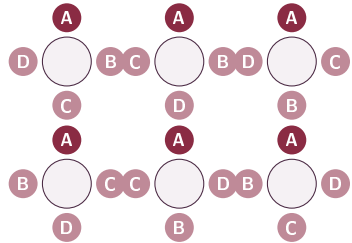
- **Way 1:** each circular permutation gives exactly $n$ linear permutations (its $n$ rotations). So we divide $n!$ by $n$.

# Number of circular permutations

**Theorem (Number of circular permutations)**

For $n \geq 1$, the number of circular permutations of $n$ distinct elements is $\frac{n!}{n} = (n-1)!$.

- **Way 1:** each circular permutation gives exactly $n$ linear permutations (its $n$ rotations). So we divide $n!$ by $n$.
- **Way 2:** the rotations do not matter, so we can **fix** the seat of $A$. Then we place the $n-1$ others in the $n-1$ free seats, as in a line: $(n-1)!$.




$A$ fixed at the top:
$3! = 6$ circular permutations of $A, B, C, D$

# Exercise

**✎ Exercise**

8 people sit around a round table (rotations are not counted).

- In how many ways can they sit?
- In how many ways can they sit if two friends $A$ and $B$ want to sit next to each other?

# Solution

**Solution**

- Circular permutations of $8$ people: $(8-1)! = 7! = 5040$.
- Glue $A$ and $B$ together as one block. We now have $7$ objects around the table: $(7-1)! = 6! = 720$ ways. Inside the block, $A$ and $B$ can be in $2$ orders ($AB$ or $BA$).
  Total: $2 \times 6! = 1440$.

See also Worksheet 1, Exercises 10 and 11 (round table).

3.2

# Combinations, Binomial Theorem & Pascal's Triangle

Unordered selections: subsets

# Combinations

**Definition**

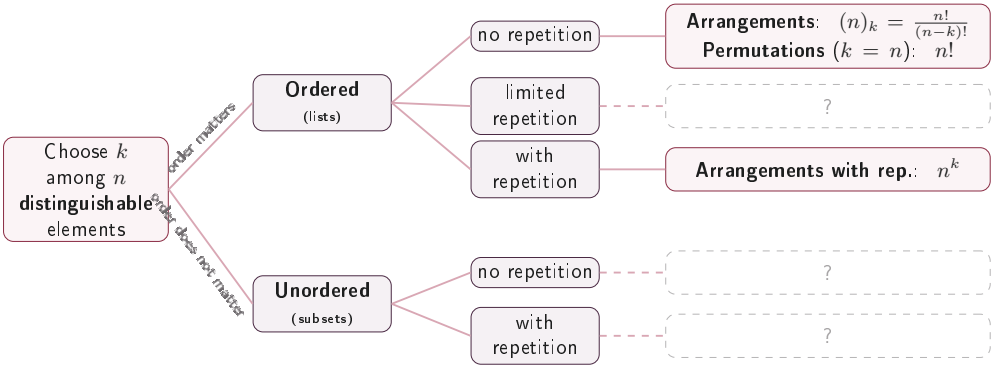
A **combination** is a selection of a subset from a finite set, where the order of selection does not matter. Simply put, a combination of a set is a **subset** of that set.

**Examples**

- $\{1, 2, 4\}$ and $\{5, 4, 1\}$ are combinations of size 3 from $\{1, 2, 3, 4, 5\}$.
- $\{A, B, C\}$ and $\{B, A, C\}$ are the **same** combination from $\{A, B, C, D\}$: the order does not matter.

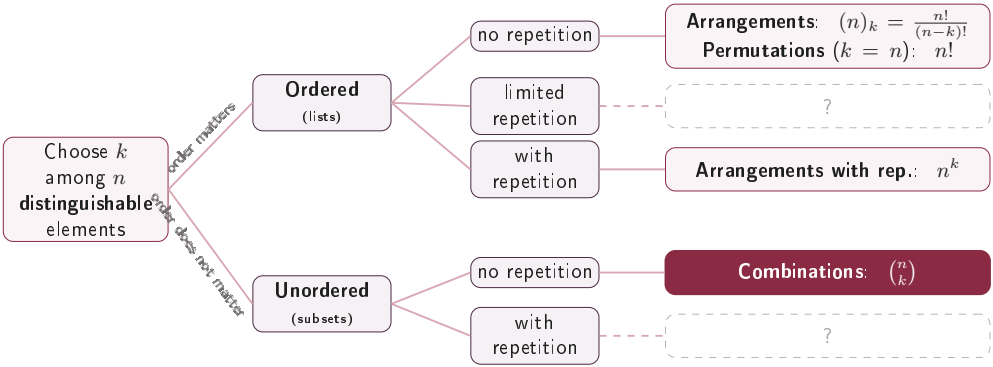
# Where are we? Combinations

We now forget the order: **unordered** selections **without repetition**, the combinations.



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We now forget the order: **unordered** selections **without repetition**, the combinations.



# $k$-combinations without repetition

**Definition**

A **$k$-combination without repetition** from a set $S$ of $n$ elements is a selection of $k$ elements of $S$ where order does not matter: a **subset** of $k$ elements.

**Examples**

- The 2-combinations of $\{A, B, C, D\}$:
  $\{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}$, so 6.
- All the combinations of $S = \{1, 2, 3\}$:
  - 0: $\{\}$ 1: $\{1\}, \{2\}, \{3\}$ 2: $\{1, 2\}, \{1, 3\}, \{2, 3\}$ 3: $\{1, 2, 3\}$

  so 8 combinations in total: all the subsets of $S$, including the empty set.

# Number of $k$-combinations without repetition

**Theorem (Number of $k$-combinations without repetition)**

For $0 \leq k \leq n$, the number of $k$-combinations without repetition from a set of $n$ elements, denoted $\binom{n}{k}$ (also written $C_n^k$), is

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!\,(n-k)!}.$$

The numbers $\binom{n}{k}$ are called **binomial coefficients**.

✎ **Proof on the board: double counting**

# Why do we divide by $k!$?

$\binom{4}{3} = 4$ **subsets**

$\{A, B, C\}$

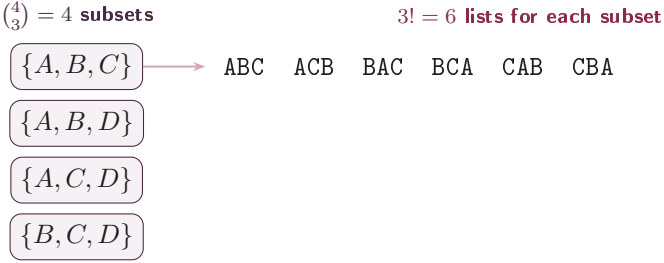
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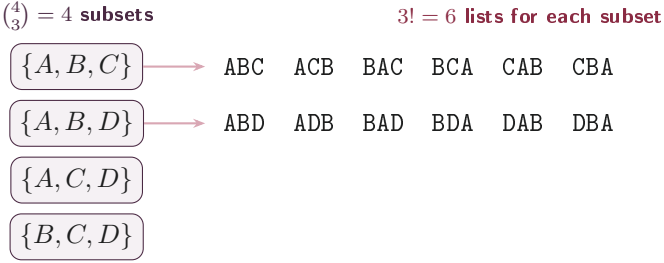
$\{B, C, D\}$

$3! = 6$ **lists for each subset**

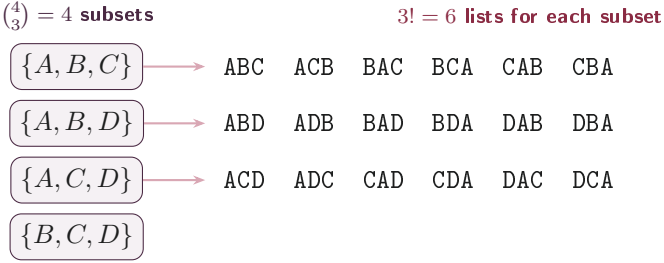
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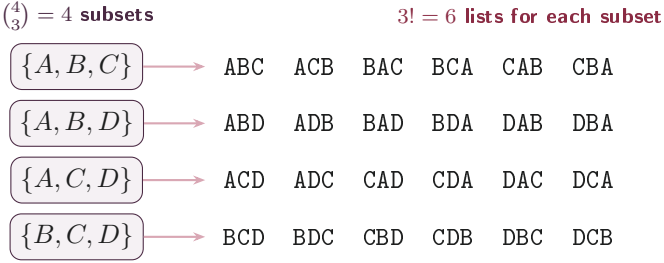
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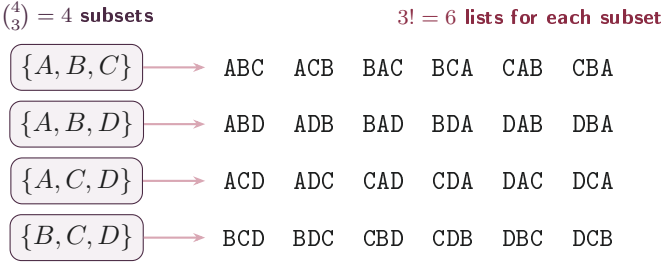
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$$(4)_3 = 24 = 3! \times \binom{4}{3} = 6 \times 4, \quad \text{so} \quad \binom{n}{k} = \frac{(n)_k}{k!}$$

# Exercise

**Exercise**

- How many ways can we select a team of 3 people from a group of 5 persons $S = \{P1, P2, P3, P4, P5\}$? List all the possible teams.
- From a group of 10 people (6 boys and 4 girls), how many ways can we select a team containing
  - 4 people?
  - 2 boys and 2 girls?

# Solution

**Solution**

- $\binom{5}{3} = \frac{5!}{3!\,2!} = \frac{5 \times 4}{2!} = 10$ teams:

  $\{P1, P2, P3\}, \{P1, P2, P4\}, \{P1, P2, P5\}, \{P1, P3, P4\}, \{P1, P3, P5\},$

  $\{P1, P4, P5\}, \{P2, P3, P4\}, \{P2, P3, P5\}, \{P2, P4, P5\}, \{P3, P4, P5\}.$

- 4 people: $\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210.$
- 2 boys and 2 girls: $\binom{6}{2} \times \binom{4}{2} = 15 \times 6 = 90.$

# Pascal's triangle

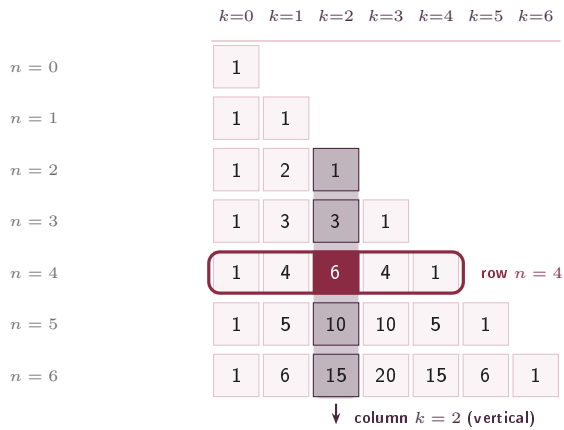
Arrange the binomial coefficients $\binom{n}{k}$ with $n$ = row number and $k$ = position in the row (column number): this is the **binomial coefficients triangle**, or **Pascal's triangle**.

$$\binom{n}{0} = 1 \quad \text{and} \quad \binom{n}{n} = 1.$$

| $n \backslash k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | |
| 2 | 1 | 2 | 1 | | | | | | | |
| 3 | 1 | 3 | 3 | 1 | | | | | | |
| 4 | 1 | 4 | 6 | 4 | 1 | | | | | |
| 5 | 1 | 5 | 10 | 10 | 5 | 1 | | | | |
| 6 | 1 | 6 | 15 | 20 | 15 | 6 | 1 | | | |
| 7 | 1 | 7 | 21 | 35 | 35 | 21 | 7 | 1 | | |
| 8 | 1 | 8 | 28 | 56 | 70 | 56 | 28 | 8 | 1 | |
| 9 | 1 | 9 | 36 | 84 | 126 | 126 | 84 | 36 | 9 | 1 |

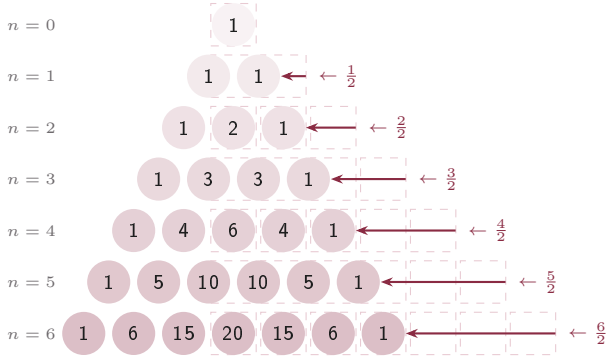
# From the table to the triangle

- **Table**: row $n$ is **horizontal**, column $k$ is **vertical**.



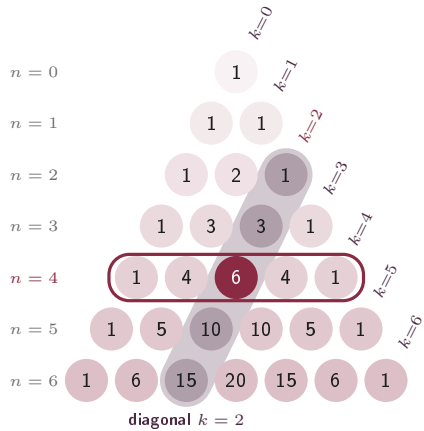
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- We **slide** row $n$ to the left by $\frac{n}{2}$ steps: every row is **centred**. The numbers do not change.



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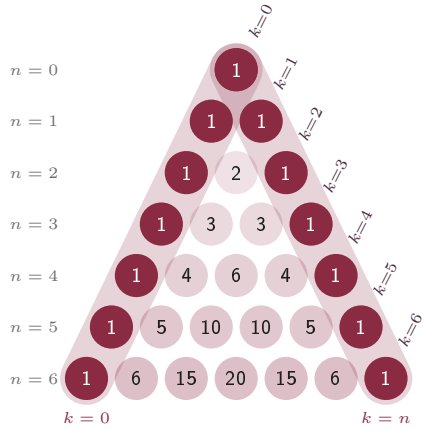



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- **Triangle**: row $n$ stays horizontal, column $k$ becomes a **diagonal** $\swarrow$, with $k$ written at its top.
- $k = 0$ gives the **left border**, $k = n$ gives the **right border**: all equal to $1$.

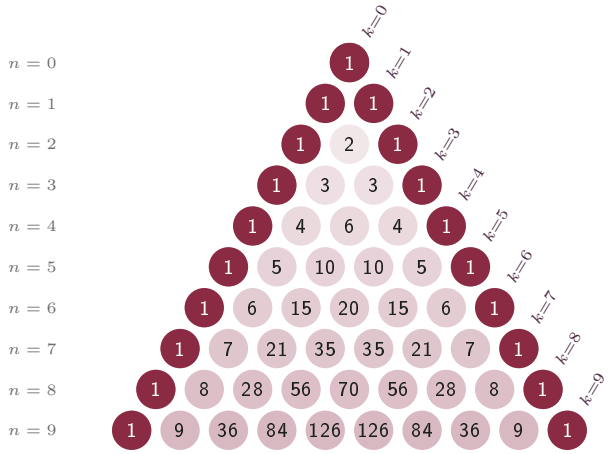
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# Pascal's triangle

Arrange the binomial coefficients $\binom{n}{k}$ with $n$ = row number and $k$ = position in the row (column number): this is the **binomial coefficients triangle**, or **Pascal's triangle**.

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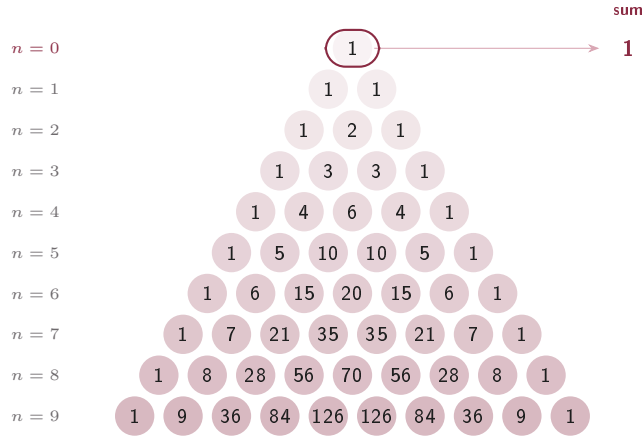


# Pascal's triangle

**Rows** $\rightarrow$ powers of 2:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 0$ $(k = 0)$:
$1 = 1 = 2^0$

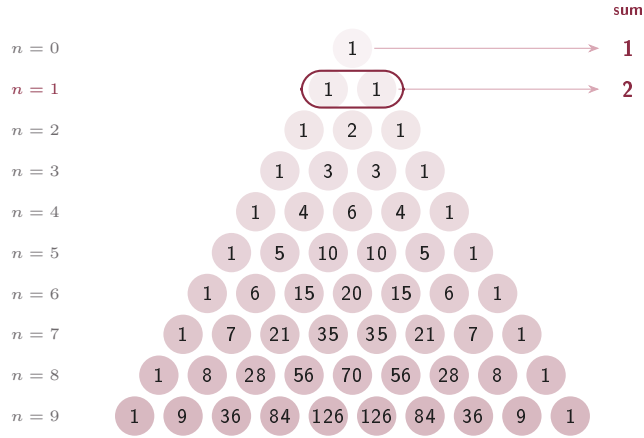


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row $n = 1$ $(k = 0, \ldots, 1)$:
$1 + 1 = 2 = 2^1$

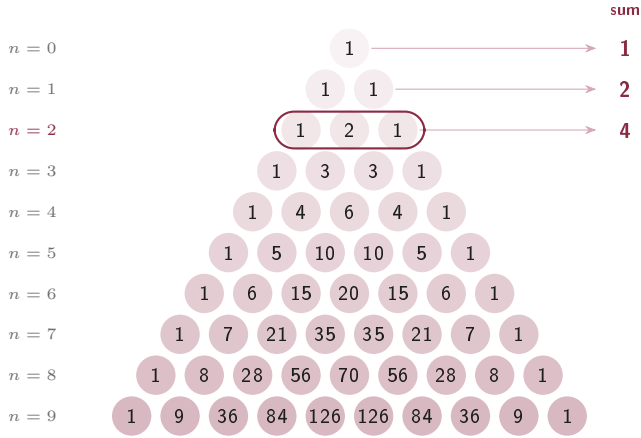


# Pascal's triangle

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$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 2$ $(k = 0, \ldots, 2)$:
$1 + 2 + 1 = 4 = 2^2$

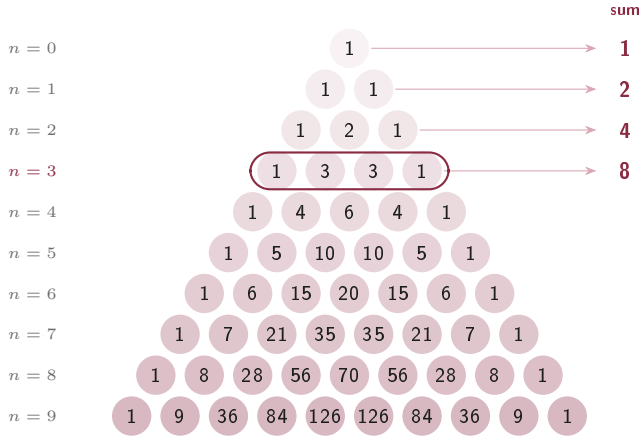


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**Rows** $\rightarrow$ powers of $2$:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 3$ $(k = 0, \ldots, 3)$:
$1 + 3 + 3 + 1 = 8 = 2^3$

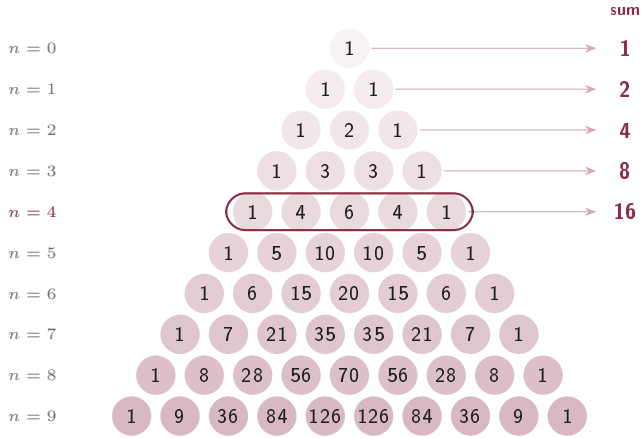


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$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 4$ $(k = 0, \ldots, 4)$:
$1 + 4 + 6 + 4 + 1 = 16 = 2^4$

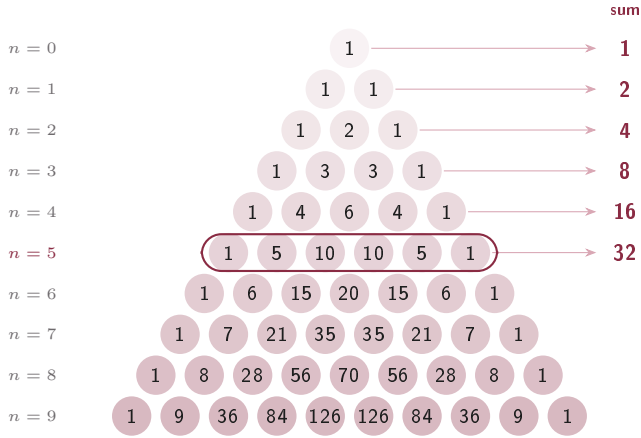


# Pascal's triangle

**Rows** $\rightarrow$ powers of 2:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 5$ $(k = 0, \ldots, 5)$:
$1+5+10+10+5+1 = 32 = 2^5$

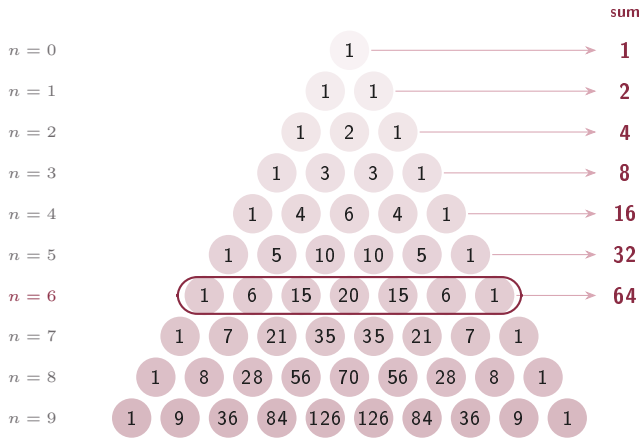


# Pascal's triangle

**Rows** $\rightarrow$ powers of $2$:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 6$ $(k = 0, \ldots, 6)$:
$1 + 6 + 15 + 20 + 15 + 6 + 1 = 64 = 2^6$

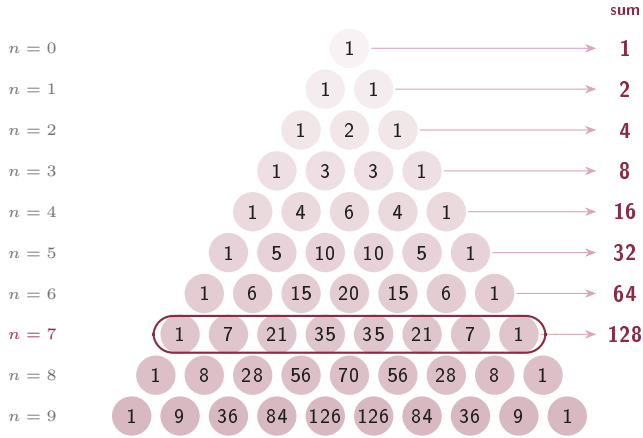


# Pascal's triangle

**Rows** $\rightarrow$ powers of $2$:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 7$ $(k = 0, \ldots, 7)$:
$1+7+21+35+35+21+7+1 = 128 = 2^7$

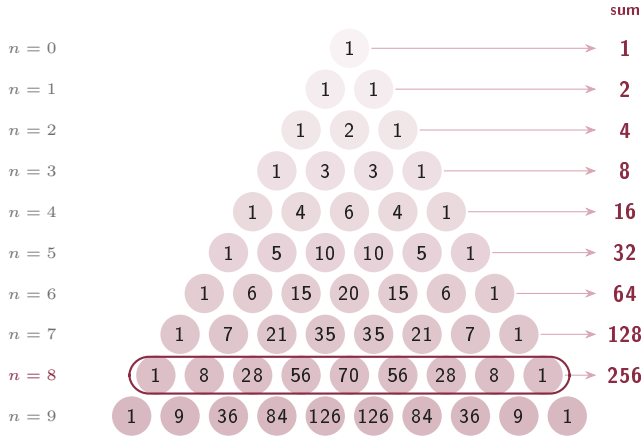


# Pascal's triangle

**Rows** $\rightarrow$ powers of 2:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 8$ $(k = 0, \ldots, 8)$:
$1 + 8 + 28 + 56 + 70 + 56 + 28 + 8 + 1 = 256 = 2^8$

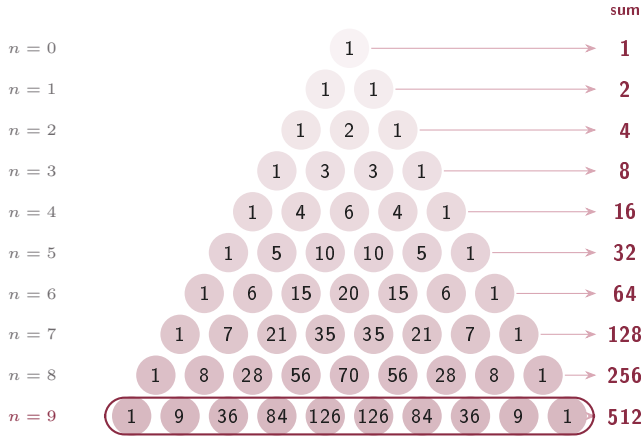


# Pascal's triangle

**Rows** $\rightarrow$ powers of $2$:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

row $n = 9$ $(k = 0, \ldots, 9)$:
$1 + 9 + 36 + 84 + 126 + 126 + 84 + 36 + 9 + 1 = 512 = 2^9$



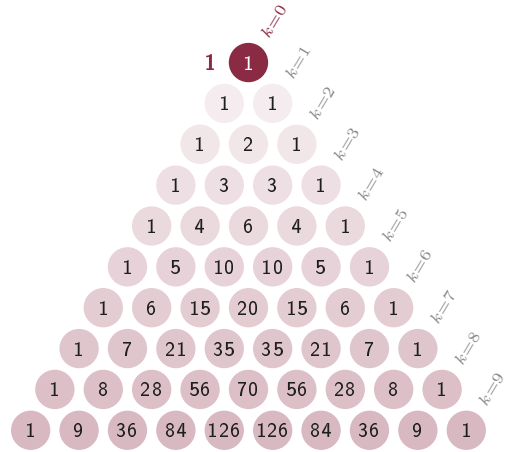
# Pascal's triangle

**Shallow diagonals** ↗
Fibonacci numbers:

$$\sum_{j\geq 0} \binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 0$:
$\binom{0}{0}$
$= 1 = 1$



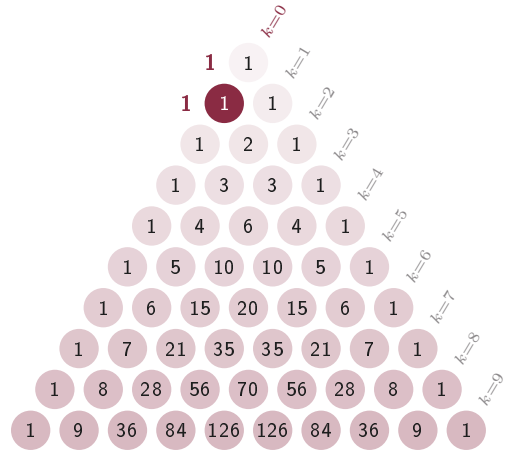
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term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 1$:
$\binom{1}{0}$
$= 1 = 1$



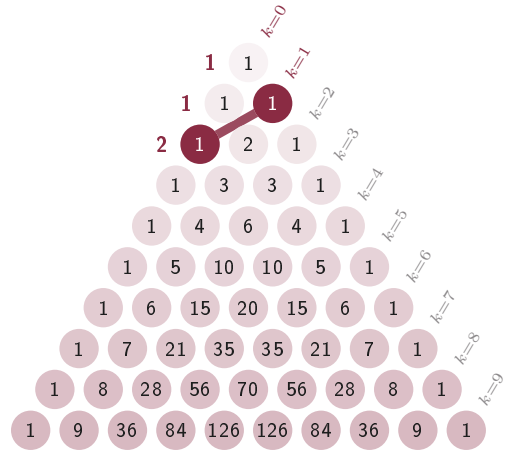
# Pascal's triangle

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term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 2$:
$\binom{2}{0} + \binom{1}{1}$
$= 1 + 1 = 2$



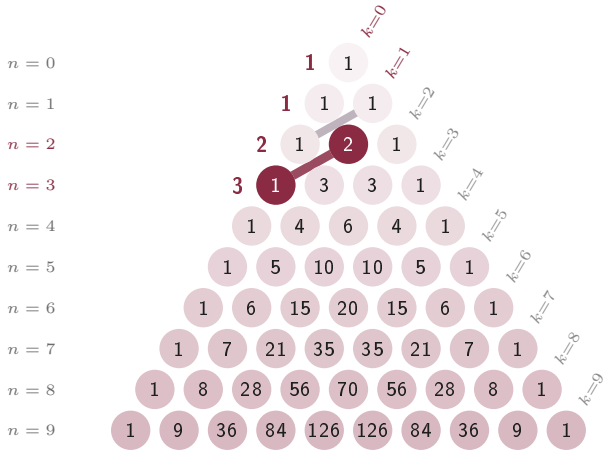
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$$\sum_{j\geq 0}\binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 3$:
$\binom{3}{0} + \binom{2}{1}$
$= 1 + 2 = 3$



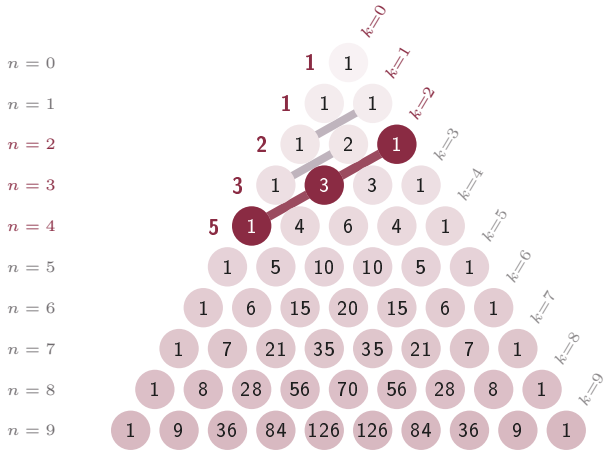
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**Shallow diagonals** ↗
Fibonacci numbers:

$$\sum_{j\geq 0} \binom{m-j}{j} = F_{m+1}$$

term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 4$:
$\binom{4}{0} + \binom{3}{1} + \binom{2}{2}$
$= 1 + 3 + 1 = 5$



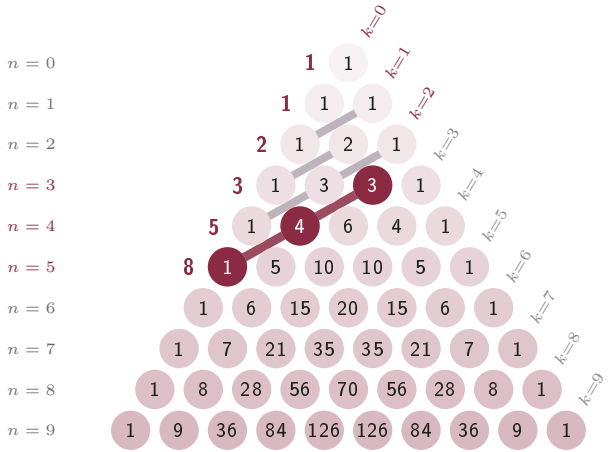
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diagonal $m = 5$:
$\binom{5}{0} + \binom{4}{1} + \binom{3}{2}$
$= 1 + 4 + 3 = 8$



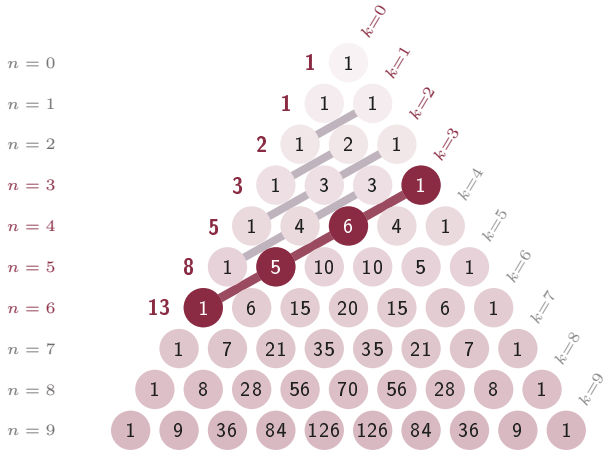
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 6$:
$\binom{6}{0} + \binom{5}{1} + \binom{4}{2} + \binom{3}{3}$
$= 1 + 5 + 6 + 1 = 13$



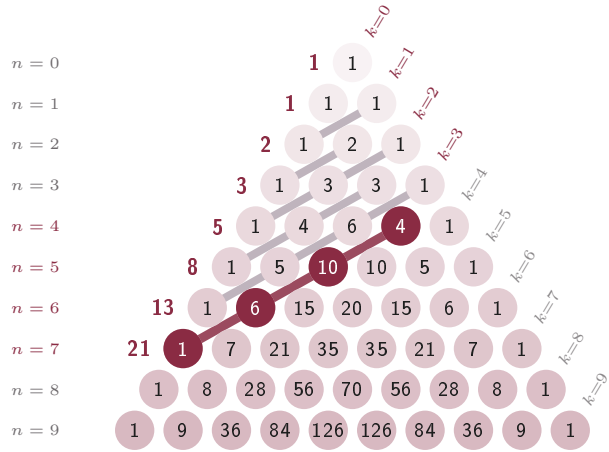
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 7$:
$\binom{7}{0} + \binom{6}{1} + \binom{5}{2} + \binom{4}{3}$
$= 1 + 6 + 10 + 4 = 21$



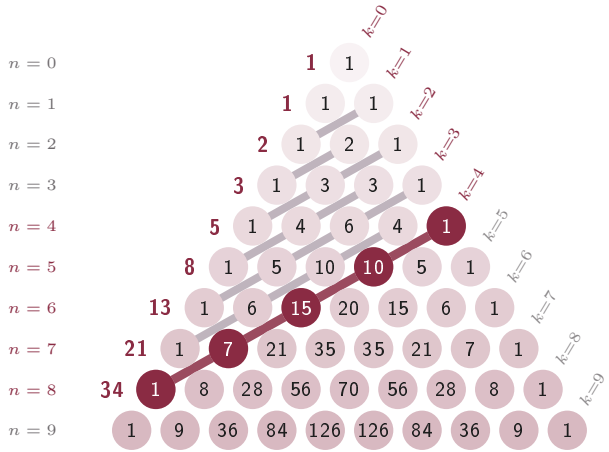
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term $\binom{m-j}{j}$: row $n = m - j$,
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diagonal $m = 8$:
$\binom{8}{0} + \binom{7}{1} + \binom{6}{2} + \binom{5}{3} + \binom{4}{4}$
$= 1 + 7 + 15 + 10 + 1 = 34$



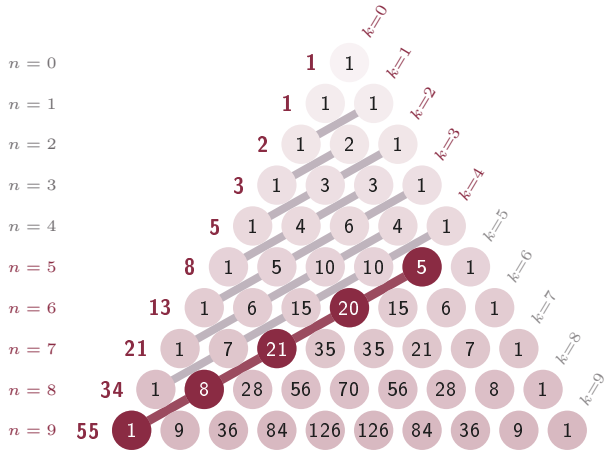
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Fibonacci numbers:

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term $\binom{m-j}{j}$: row $n = m - j$,
diagonal $k = j$.

diagonal $m = 9$:
$\binom{9}{0} + \binom{8}{1} + \binom{7}{2} + \binom{6}{3} + \binom{5}{4}$
$= 1 + 8 + 21 + 20 + 5 = 55$



# Symmetry

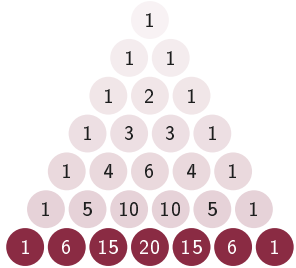
**Theorem (Symmetry Property)**

For $0 \leq k \leq n$: $\binom{n}{k} = \binom{n}{n-k}$.

✎ **Algebraic and combinatorial proofs on the board**

In Pascal's triangle, every row reads the same from left to right and from right to left:

1 6 15 20 15 6 1



# Symmetry

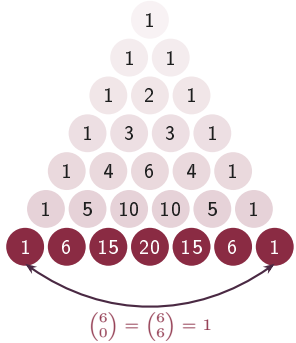
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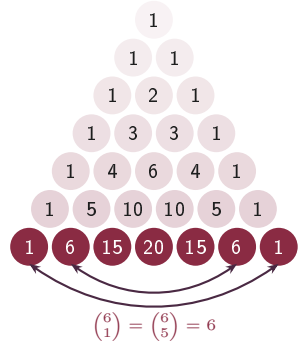
**Theorem (Symmetry Property)**

For $0 \leq k \leq n$: $\displaystyle\binom{n}{k} = \binom{n}{n-k}$.

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# Symmetry

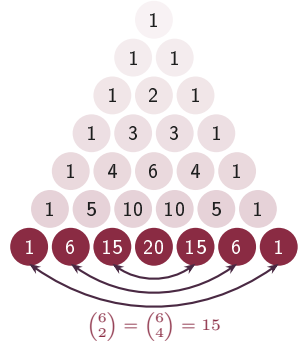
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For $0 \leq k \leq n$: $$\binom{n}{k} = \binom{n}{n-k}.$$

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# Symmetry

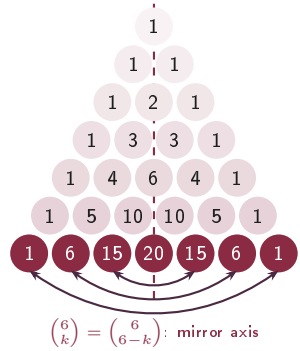
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# Pascal's rule

**Theorem (Recurrence Relation (Pascal's rule))**

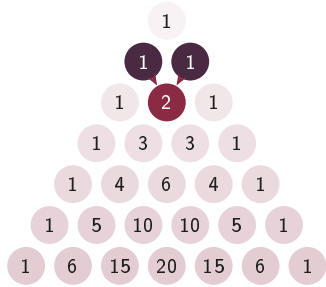
For $n \geq 1$ and $1 \leq k \leq n-1$:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

✎ **Proof on the board**

**Remark**

For now, if $k > n$, $k < 0$ or $n < 0$, we set $\binom{n}{k} = 0$.




$\binom{2}{1} = \binom{1}{0} + \binom{1}{1}$: $2 = 1 + 1$

# Pascal's rule

**Theorem (Recurrence Relation (Pascal's rule))**

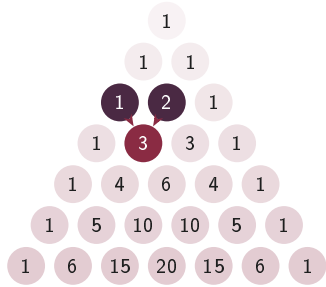
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# Pascal's rule

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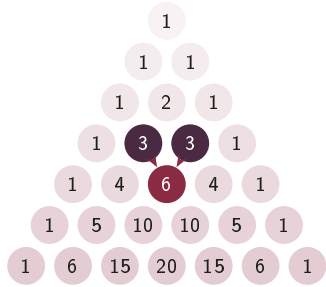
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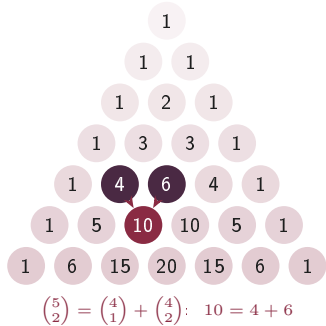
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✎ **Proof on the board**

**Remark**

For now, if $k > n$, $k < 0$ or $n < 0$, we set $\binom{n}{k} = 0$.




$\binom{5}{2} = \binom{4}{1} + \binom{4}{2}$: $10 = 4 + 6$

# Pascal's rule

**Theorem (Recurrence Relation (Pascal's rule))**

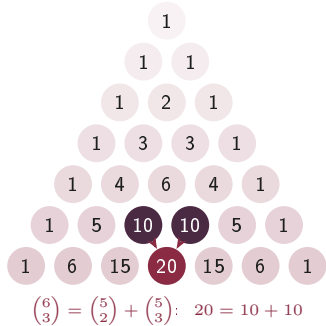
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✎ **Proof on the board**

**Remark**

For now, if $k > n$, $k < 0$ or $n < 0$, we set $\binom{n}{k} = 0$.



# Vandermonde's identity

**Theorem (Vandermonde's Identity)**

For any non-negative integers $m$, $n$ and $k$:

$$\binom{m+n}{k} = \sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r}.$$

✎ **Proof on the board: double counting**

# Worksheet 1 – Exercise 20 (lecture)

✎ **Exercise (Worksheet 1, Exercise 20)**

Prove in two different ways the following identity (known as the absorption/extraction identity), for $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

# Worksheet 1 – Exercise 20 (lecture)

**✎ Exercise (Worksheet 1, Exercise 20)**

Prove in two different ways the following identity (known as the absorption/extraction identity), for $1 \leq k \leq n$:

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

**Two ways**

- **Way 1 (algebra):** use the formula $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$.
- **Way 2 (counting):** find one question whose answer is the left side *and* the right side.

# Exercise 20: first way (with the formula)

**Solution (Exercise 20 – algebraic proof)**

We start from the left side and use $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$:

$$k\binom{n}{k} = k \cdot \frac{n!}{k!\,(n-k)!} \qquad \text{(definition of } \tbinom{n}{k}\text{)}$$

# Exercise 20: first way (with the formula)

**Solution (Exercise 20 – algebraic proof)**

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$$= \frac{n!}{(k-1)!\,(n-k)!} \qquad \text{because } \frac{k}{k!} = \frac{1}{(k-1)!}$$

# Exercise 20: first way (with the formula)

**Solution (Exercise 20 – algebraic proof)**

We start from the left side and use $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$:

$$
\begin{aligned}
k\binom{n}{k} &= k \cdot \frac{n!}{k!\,(n-k)!} && \text{(definition of } \tbinom{n}{k}\text{)} \\
&= \frac{n!}{(k-1)!\,(n-k)!} && \text{because } \tfrac{k}{k!} = \tfrac{1}{(k-1)!} \\
&= n \cdot \frac{(n-1)!}{(k-1)!\,(n-k)!} && \text{because } n! = n \cdot (n-1)!
\end{aligned}
$$

# Exercise 20: first way (with the formula)

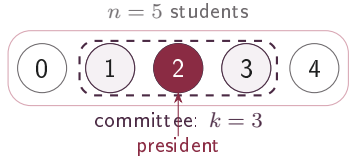
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&= \frac{n!}{(k-1)!\,(n-k)!} && \text{because } \tfrac{k}{k!} = \tfrac{1}{(k-1)!} \\
&= n \cdot \frac{(n-1)!}{(k-1)!\,(n-k)!} && \text{because } n! = n \cdot (n-1)! \\
&= n\binom{n-1}{k-1} && \text{because } (n-1) - (k-1) = n-k.
\end{aligned}
$$

# Exercise 20: second way (by counting)

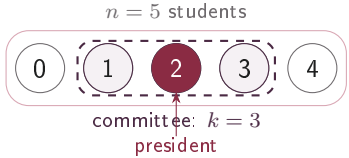
**Question:** in a group of $n$ students, we choose a **committee** of $k$ students, and **one** of them is the **president**. How many ways?



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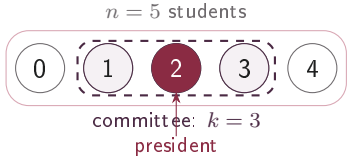
- **Way 1:** choose the committee: $\binom{n}{k}$; then the president among its $k$ members: $k$.
  Total: $k\binom{n}{k}$.



# Exercise 20: second way (by counting)

**Question:** in a group of $n$ students, we choose a **committee** of $k$ students, and **one** of them is the **president**. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the president among its $k$ members: $k$.
  Total: $k\binom{n}{k}$.
- **Way 2:** choose the president first: $n$; then the $k-1$ other members among the $n-1$ other students: $\binom{n-1}{k-1}$.
  Total: $n\binom{n-1}{k-1}$.

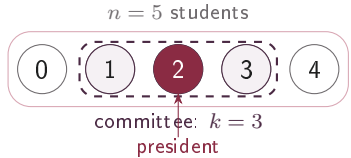


# Exercise 20: second way (by counting)

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  Total: $k\binom{n}{k}$.
- **Way 2:** choose the president first: $n$; then the $k-1$ other members among the $n-1$ other students: $\binom{n-1}{k-1}$.
  Total: $n\binom{n-1}{k-1}$.

We counted the same thing twice, so the two numbers are **equal**.




check: $3 \cdot \binom{5}{3} = 3 \cdot 10 = 30$
$5 \cdot \binom{4}{2} = 5 \cdot 6 = 30$

# Worksheet 1 – Exercise 21 (lecture)

**✎ Exercise (Worksheet 1, Exercise 21)**

Establish the "committee-subcommittee" identity, for $0 \leq r \leq k \leq n$:

$$\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}.$$

# Worksheet 1 – Exercise 21 (lecture)

✎ **Exercise (Worksheet 1, Exercise 21)**

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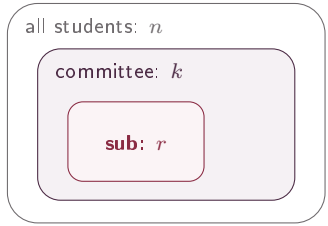
$$\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}.$$

**Idea**

The name of the identity gives the question: choose a **committee**, then a **sub-committee** inside it.

# Exercise 21: first way (by counting)

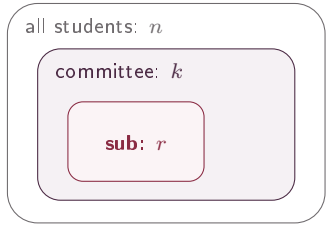
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# Exercise 21: first way (by counting)

**Question:** from $n$ students, we choose a **committee** of $k$ students, and inside it a **subcommittee** of $r$ students. How many ways?

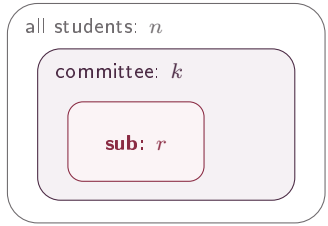
- **Way 1:** choose the committee: $\binom{n}{k}$; then the subcommittee inside it: $\binom{k}{r}$.
  Total: $\binom{n}{k}\binom{k}{r}$.



# Exercise 21: first way (by counting)

**Question:** from $n$ students, we choose a **committee** of $k$ students, and inside it a **subcommittee** of $r$ students. How many ways?

- **Way 1:** choose the committee: $\binom{n}{k}$; then the subcommittee inside it: $\binom{k}{r}$.
  Total: $\binom{n}{k}\binom{k}{r}$.
- **Way 2:** choose the subcommittee first: $\binom{n}{r}$; then the $k-r$ other committee members among the $n-r$ remaining students: $\binom{n-r}{k-r}$.
  Total: $\binom{n}{r}\binom{n-r}{k-r}$.

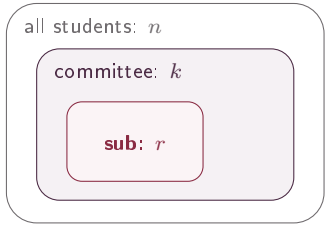


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  Total: $\binom{n}{r}\binom{n-r}{k-r}$.

Same question, so the two numbers are **equal**.




check ($n=5$, $k=3$, $r=2$):
$\binom{5}{3}\binom{3}{2} = 10 \cdot 3 = 30$
$\binom{5}{2}\binom{3}{1} = 10 \cdot 3 = 30$

# Exercise 21: second way (with the formula)

**Solution (Exercise 21 – algebraic proof)**

**Left side:**

$$\binom{n}{k}\binom{k}{r} = \frac{n!}{k!\,(n-k)!} \cdot \frac{k!}{r!\,(k-r)!}$$

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($k!$ is on top and at the bottom: it cancels.)

# Exercise 21: second way (with the formula)

**Solution (Exercise 21 – algebraic proof)**

**Left side:**

$$\binom{n}{k}\binom{k}{r} = \frac{n!}{k!\,(n-k)!} \cdot \frac{k!}{r!\,(k-r)!} = \frac{n!}{r!\,(k-r)!\,(n-k)!}$$

($k!$ is on top and at the bottom: it cancels.)

**Right side:**

$$\binom{n}{r}\binom{n-r}{k-r} = \frac{n!}{r!\,(n-r)!} \cdot \frac{(n-r)!}{(k-r)!\,(n-k)!} = \frac{n!}{r!\,(k-r)!\,(n-k)!}$$

($(n-r)!$ cancels, and $(n-r)-(k-r) = n-k$.)

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($(n-r)!$ cancels, and $(n-r)-(k-r) = n-k$.)

The two sides give the **same** fraction, so the identity is true.

# The Binomial Theorem

**Theorem (The Binomial Theorem)**

For any real numbers $a$, $b$ and every non-negative integer $n$:

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{n}b^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k.$$

✎ Combinatorial proof on the board; induction as exercise

**Corollary**

For every $n \geq 1$: $\displaystyle\sum_{k=0}^{n}\binom{n}{k} = 2^n$ and $\displaystyle\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0.$

# The binomial theorem: a picture

$$(a+b)^3 = \boxed{(a+b)}\boxed{(a+b)}\boxed{(a+b)}$$

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choose $a$ or $b$ in each factor: $2^3 = 8$ words

| | | | |
|---|---|---|---|
| aaa | aab<br>aba<br>baa | abb<br>bab<br>bba | bbb |

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|---|---|---|---|
| 0 $b$: $\binom{3}{0} = 1$ | 1 $b$: $\binom{3}{1} = 3$ | 2 $b$: $\binom{3}{2} = 3$ | 3 $b$: $\binom{3}{3} = 1$ |

# The binomial theorem: a picture

$$(a+b)^3 = \boxed{(a+b)}\boxed{(a+b)}\boxed{(a+b)}$$

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| 0 $b$: $\binom{3}{0} = 1$ | 1 $b$: $\binom{3}{1} = 3$ | 2 $b$: $\binom{3}{2} = 3$ | 3 $b$: $\binom{3}{3} = 1$ |

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1 3 3 1 = row 3 of Pascal's triangle)

# Solution: The Binomial Theorem by induction

**Solution (proof by induction on $n$)**

For $n = 0$ both sides equal $1$. Assume the formula for $n$. Then

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k = \sum_{k=0}^{n}\binom{n}{k}a^{n+1-k}b^k + \sum_{k=1}^{n+1}\binom{n}{k-1}a^{n+1-k}b^k,$$

(we put $k \to k-1$ in the second sum). With the convention $\binom{n}{-1} = \binom{n}{n+1} = 0$ and Pascal's rule $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$:

$$(a+b)^{n+1} = \sum_{k=0}^{n+1}\binom{n+1}{k}a^{n+1-k}b^k.$$

**Corollary**: take $a = b = 1$, then $a = 1$, $b = -1$ (for $n \geq 1$).

# Exercise

**✎ Exercise**

Determine the coefficient of $a^{17}b^9$ in the expansion of $(a+b)^{26}$.

# Solution

**Solution**

The general term is $\binom{26}{k} a^{26-k} b^k$. We need $26-k=17$, so $k=9$. The coefficient is

$$\binom{26}{9} = 3\,124\,550.$$

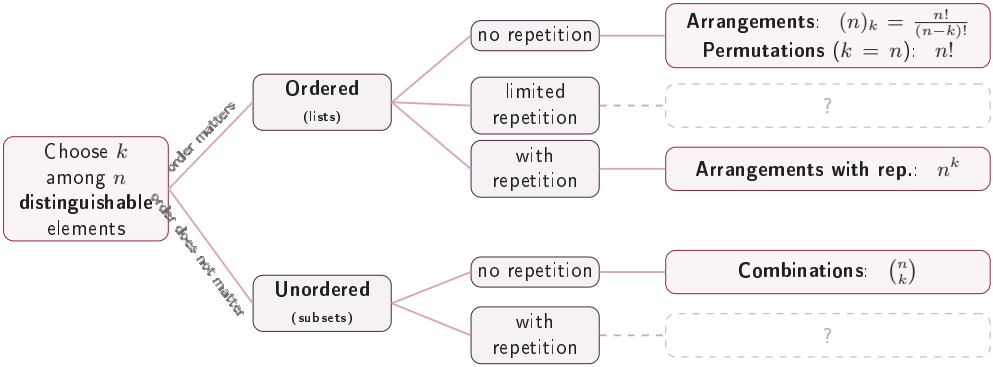
3.3

# Permutations with Limited Repetition & Multinomials

The MISSISSIPPI problem

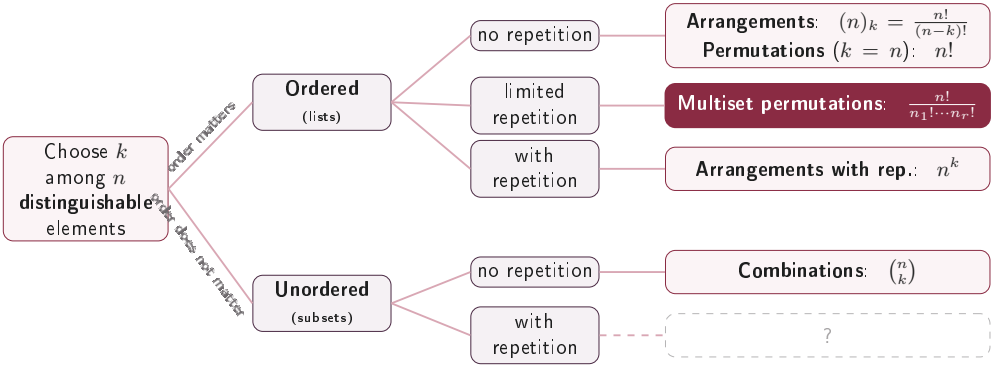
# Where are we? Permutations of a multiset

Ordered again, each element used a **limited** number of times (MISSISSIPPI).



# Where are we? Permutations of a multiset

Ordered again, each element used a **limited** number of times (MISSISSIPPI).



# The MISSISSIPPI problem




How many distinct words can be formed by rearranging the letters of "MISSISSIPPI"?

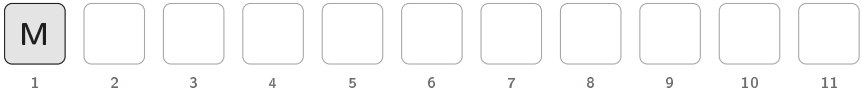
- Not $11!$: permuting identical letters gives the **same** word.
- Choose the **positions** of each letter instead:

$$\underbrace{\binom{11}{1}}_{M}\underbrace{\binom{10}{4}}_{I}\underbrace{\binom{6}{4}}_{S}\underbrace{\binom{2}{2}}_{P} = \frac{11!}{1!\,10!}\cdot\frac{10!}{4!\,6!}\cdot\frac{6!}{4!\,2!}\cdot\frac{2!}{2!} = \frac{11!}{1!\,4!\,4!\,2!} = 34\,650.$$

# MISSISSIPPI: choosing the positions

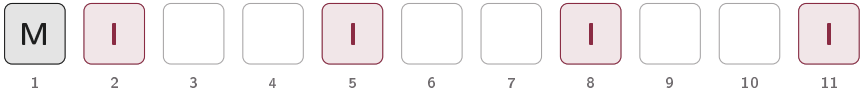


# MISSISSIPPI: choosing the positions




positions of the M's: choose 1 among the 11 free positions: $\binom{11}{1} = 11$

# MISSISSIPPI: choosing the positions




positions of the I's: choose $4$ among the $10$ free positions: $\binom{10}{4} = 210$

# MISSISSIPPI: choosing the positions




positions of the S's: choose $4$ among the $6$ free positions: $\binom{6}{4} = 15$

# MISSISSIPPI: choosing the positions




positions of the P's: choose 2 among the 2 free positions: $\binom{2}{2} = 1$

# MISSISSIPPI: choosing the positions



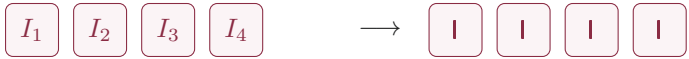

$$\binom{11}{1}\binom{10}{4}\binom{6}{4}\binom{2}{2} = 11 \times 210 \times 15 \times 1 = 34\,650$$

# Another way to see it

**Double counting**

Label the letters $I_1, \ldots, I_4$, $S_1, \ldots, S_4$, $P_1, P_2$. There are $11!$ arrangements of these 11 **distinct** letters, and each word is obtained exactly $4!\,4!\,2!$ times (by permuting the labels). Hence there are

$$\frac{11!}{4!\,4!\,2!} \text{ words.}$$




$4!$ labelled versions give the same word

# Permutations with limited repetition

**Theorem (Permutations with Limited Repetition)**

Let $S$ be a collection of $k$ distinct elements $e_1, \ldots, e_k$, where $e_i$ is repeated $n_i$ times, with $n_1 + n_2 + \cdots + n_k = n$. The number of permutations of all the elements of $S$ is

$$\binom{n}{n_1, n_2, \ldots, n_k} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-\sum_{i=1}^{k-1} n_i}{n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}.$$

The numbers $\binom{n}{n_1,\ldots,n_k}$ are called **multinomial coefficients**.

✎ Proof on the board

# Multisets

## Remarks

- The collection $S$ is denoted $\{n_1 \cdot e_1,\ n_2 \cdot e_2,\ \ldots,\ n_k \cdot e_k\}$ and is called a **multiset**: unlike a set, it can contain the same element several times.
- Infinite repetition is denoted by $\infty$: in $\{\infty \cdot a,\ 5 \cdot b,\ 3 \cdot c\}$, the element $a$ appears infinitely often, $b$ five times and $c$ three times.
- Problems with infinite repetition are generally simpler than those with finite repetition. For $A = \{\infty \cdot a_1, \ldots, \infty \cdot a_n\}$, the number of arrangements of length $k$ is

$$n^k,$$

which are exactly the $k$-permutations with repetition.

# Properties of multinomial coefficients

Multinomial coefficients count the ways to divide $n$ distinct objects into $k$ **labelled** (distinguishable) groups of sizes $n_1, \ldots, n_k$, with $n_1 + \cdots + n_k = n$.

**Theorem (Recursive Formula for Multinomial Coefficients)**

Let $n \geq 1$, $k \geq 2$, and $n_1, \ldots, n_k \geq 0$ with $n_1 + \cdots + n_k = n$. Then

$$\binom{n}{n_1, \ldots, n_k} = \sum_{i=1}^{k} \binom{n-1}{n_1, \ldots, n_i - 1, \ldots, n_k}$$

(a term with a negative entry $n_i - 1 < 0$ is taken to be $0$; for $k = 2$ this is Pascal's rule).

✎ **Exercise.** Prove the theorem both algebraically and combinatorially.

# Solution: Recursive formula for multinomial coefficients

## Solution

**Combinatorial proof.** $\binom{n}{n_1,\ldots,n_k}$ counts the words of length $n$ containing $n_i$ letters $e_i$. Classify these words by their *first* letter. If the first letter is $e_i$ (possible only if $n_i \geq 1$), the remaining $n-1$ letters form a word with $n_i - 1$ letters $e_i$ and $n_j$ letters $e_j$ for $j \neq i$: there are $\binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k}$ of them. The cases are disjoint; add them.

**Algebraic proof.** For $n_i \geq 1$,

$$\binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k} = \frac{(n-1)!}{n_1!\cdots(n_i-1)!\cdots n_k!} = \frac{(n-1)!}{n_1!\cdots n_k!}\, n_i,$$

and this is also true (both sides are $0$) when $n_i = 0$. Summing over $i$:

$$\sum_{i=1}^{k} \binom{n-1}{n_1,\ldots,n_i-1,\ldots,n_k} = \frac{(n-1)!}{n_1!\cdots n_k!}\,(n_1+\cdots+n_k) = \frac{n!}{n_1!\cdots n_k!}.$$

# The Multinomial Theorem

**Theorem (Multinomial Theorem)**

For any positive integer $n$ and any real numbers $x_1, \ldots, x_k$:

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{\substack{n_1+\cdots+n_k=n \\ n_1,\ldots,n_k \geq 0}} \binom{n}{n_1, n_2, \ldots, n_k} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}.$$

✎ **Exercise**

- Give a combinatorial proof, using arguments similar to the binomial theorem.
- Prove the multinomial theorem by induction on $n$, and by induction on $k$.

# Solution: Multinomial Theorem, combinatorial proof

**Solution (combinatorial proof)**

Write $(x_1 + x_2 + \cdots + x_k)^n$ as a product of $n$ identical factors. Expanding, each term corresponds to choosing one of $x_1, \ldots, x_k$ in each of the $n$ factors, that is, to a word of length $n$ over the letters $x_1, \ldots, x_k$.
The term $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ is obtained from the words containing exactly $n_1$ letters $x_1$, ..., $n_k$ letters $x_k$. By the previous theorem, there are

$$\binom{n}{n_1, n_2, \ldots, n_k}$$

such words, which is the coefficient of $x_1^{n_1} \cdots x_k^{n_k}$.

# Solution: Multinomial Theorem by induction on $n$

**Solution (proof by induction on $n$)**

For $n = 1$ the formula is $x_1 + \cdots + x_k$. Assume it for $n$ and multiply by $(x_1 + \cdots + x_k)$. A monomial $x_1^{n_1} \cdots x_k^{n_k}$ with $n_1 + \cdots + n_k = n + 1$ is obtained from $x_i \cdot x_1^{n_1} \cdots x_i^{n_i - 1} \cdots x_k^{n_k}$, for $i = 1, \ldots, k$. Its coefficient is therefore

$$\sum_{i=1}^{k} \binom{n}{n_1, \ldots, n_i - 1, \ldots, n_k} = \binom{n+1}{n_1, \ldots, n_k}$$

by the recursive formula for multinomial coefficients.

# Solution: Multinomial Theorem by induction on $k$

**Solution (proof by induction on $k$)**

For $k = 2$ this is the binomial theorem. Assume the result for $k - 1$ variables. By the binomial theorem,

$$(x_1 + \cdots + x_k)^n = \sum_{n_k=0}^{n} \binom{n}{n_k} (x_1 + \cdots + x_{k-1})^{n-n_k} \, x_k^{n_k},$$

and by the induction hypothesis the coefficient of $x_1^{n_1} \cdots x_k^{n_k}$ is

$$\binom{n}{n_k} \binom{n - n_k}{n_1, \ldots, n_{k-1}} = \frac{n!}{n_k!\,(n - n_k)!} \cdot \frac{(n - n_k)!}{n_1! \cdots n_{k-1}!} = \binom{n}{n_1, \ldots, n_k}.$$

# Example: $(x_1 + x_2 + x_3)^3$

$$
\begin{aligned}
(x_1 + x_2 + x_3)^3 &= \sum_{\substack{n_1+n_2+n_3=3 \\ n_1,n_2,n_3 \geq 0}} \binom{3}{n_1, n_2, n_3} x_1^{n_1} x_2^{n_2} x_3^{n_3} \\
&= \binom{3}{0,0,3} x_3^3 + \binom{3}{0,3,0} x_2^3 + \binom{3}{0,1,2} x_2 x_3^2 + \binom{3}{0,2,1} x_2^2 x_3 + \binom{3}{1,1,1} x_1 x_2 x_3 \\
&\quad + \binom{3}{1,2,0} x_1 x_2^2 + \binom{3}{1,0,2} x_1 x_3^2 + \binom{3}{2,0,1} x_1^2 x_3 + \binom{3}{2,1,0} x_1^2 x_2 + \binom{3}{3,0,0} x_1^3
\end{aligned}
$$

# Example: $(x_1 + x_2 + x_3)^3$, computing the coefficients

**Computing the coefficients**

$$x_1^3 + x_2^3 + x_3^3 + 3\left(x_1^2 x_2 + x_1^2 x_3 + x_1 x_2^2 + x_2^2 x_3 + x_1 x_3^2 + x_2 x_3^2\right) + 6\, x_1 x_2 x_3.$$

There are $10$ terms: we will see that $10 = \left(\!\binom{3}{3}\!\right) = \binom{5}{3}$.

# Exercise

**✎ Exercise**

Without expanding completely, find the coefficients in $(a+b+2c)^5$ of the following terms (watch the factor 2):

- $a^2b^2c$
- $a^3bc$
- $b^5$

# Solution

**Solution**

The general term of $(a+b+2c)^5$ is

$$\binom{5}{n_1, n_2, n_3} a^{n_1} b^{n_2} (2c)^{n_3} = \binom{5}{n_1, n_2, n_3} 2^{n_3} a^{n_1} b^{n_2} c^{n_3}.$$

- $a^2b^2c$: $\dfrac{5!}{2!\,2!\,1!} \times 2 = 30 \times 2 = 60.$
- $a^3bc$: $\dfrac{5!}{3!\,1!\,1!} \times 2 = 20 \times 2 = 40.$
- $b^5$: $\dfrac{5!}{0!\,5!\,0!} \times 2^0 = 1.$

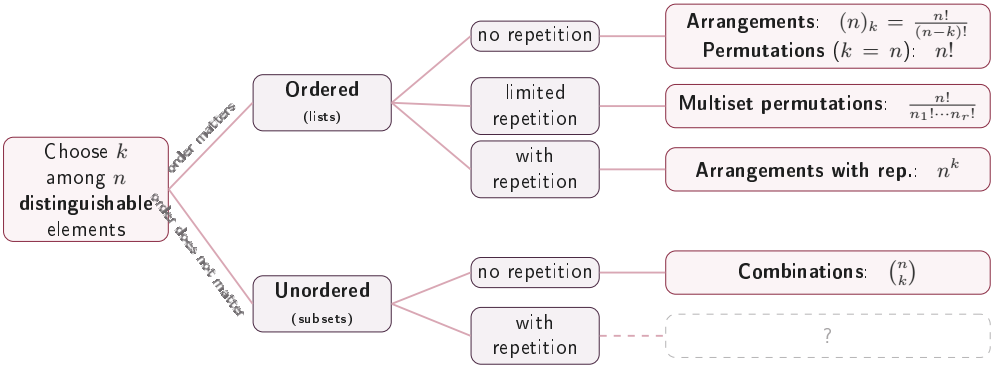
3.4

# Combinations with Repetition

Unordered selections where repetition is allowed

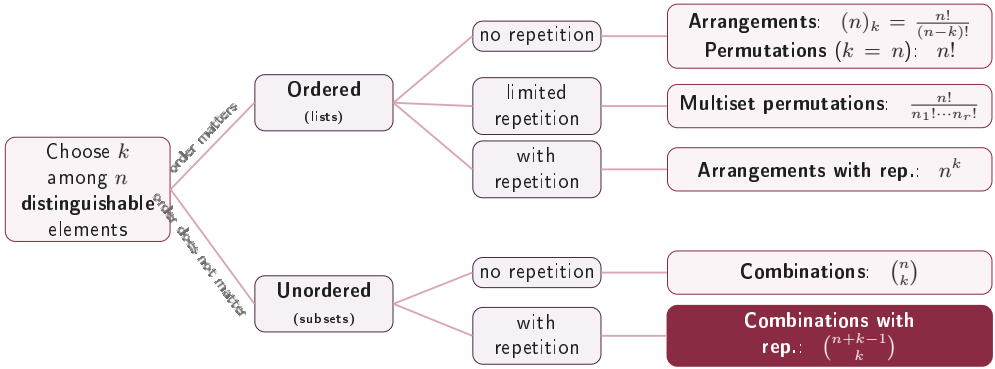
# Where are we? Combinations with repetition

Last leaf: **unordered** selections **with repetition**.



# Where are we? Combinations with repetition

Last leaf: **unordered** selections **with repetition**.



# Combinations with repetition

Let $S = \{1, 2, 3\}$. The 2-combinations of $S$:

- **without** repetition: $\{1, 2\},\ \{1, 3\},\ \{2, 3\}$;
- **with** repetition: $\{1, 1\},\ \{1, 2\},\ \{1, 3\},\ \{2, 2\},\ \{2, 3\},\ \{3, 3\}$, so 6 solutions.

*If $S = \{1, 2, \ldots, n\}$, how many $k$-combinations with repetition are there?*

**Definition**

A **$k$-combination with repetition** from $S = \{1, 2, \ldots, n\}$ is an **unordered** selection of $k$ elements, where repetition of elements is allowed.

# Combinations with repetition and multisets

**Remarks**

- If $x_1, \ldots, x_n$ are the repetitions of $e_1, \ldots, e_n$, then

$$x_1 + x_2 + \cdots + x_n = k, \qquad 0 \leq x_i \leq k.$$

- A $k$-combination with repetition is the same as a **multiset** of $k$ elements $\{x_1 \cdot e_1, \ldots, x_n \cdot e_n\}$ with $x_1 + \cdots + x_n = k$.

# Number of $k$-combinations with repetition

**Theorem (Number of $k$-combinations with repetition)**

The number of $k$-combinations with repetition from a set of $n$ elements, denoted $\left(\!\binom{n}{k}\!\right)$, is

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

✎ **Proof on the board: stars and bars**

# Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

**Back to $S = \{1, 2, 3\}$ and $k = 2$ ($3$ groups, so $2$ bars)**

$$\{1, 1\} \quad \longleftrightarrow \quad \star\star \,||$$

# Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

**Back to $S = \{1, 2, 3\}$ and $k = 2$ ($3$ groups, so $2$ bars)**

$$\{1, 1\} \quad \longleftrightarrow \quad \star\star \,||$$
$$\{1, 2\} \quad \longleftrightarrow \quad \star|\star\,|$$

# Stars and bars

$$\underbrace{\star\,\star\,\cdots\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

**Back to $S = \{1, 2, 3\}$ and $k = 2$ (3 groups, so 2 bars)**

$$\begin{aligned} \{1,1\} &\longleftrightarrow \star\star\,||\\ \{1,2\} &\longleftrightarrow \star|\star\,|\\ \{1,3\} &\longleftrightarrow \star||\star \end{aligned}$$

# Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

**Back to $S = \{1, 2, 3\}$ and $k = 2$ ($3$ groups, so $2$ bars)**

$$\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|
\end{aligned}$$

# Stars and bars

$$\underbrace{\star\,\star\,\cdots\,\star}_{x_1 \text{ stars}} \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_2 \text{ stars}} \;\Big|\; \cdots \;\Big|\; \underbrace{\star\,\star\,\cdots\,\star}_{x_n \text{ stars}} \qquad k \text{ stars, } n-1 \text{ bars}$$

**Back to $S = \{1, 2, 3\}$ and $k = 2$ ($3$ groups, so $2$ bars)**

$$\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|\\
\{2,3\} &\longleftrightarrow |\,\star\,|\star
\end{aligned}$$

# Stars and bars




$k$ stars, $n-1$ bars

**Back to $S=\{1,2,3\}$ and $k=2$ (3 groups, so 2 bars)**

$$
\begin{aligned}
\{1,1\} &\longleftrightarrow \star\star\,||\\
\{1,2\} &\longleftrightarrow \star|\star\,|\\
\{1,3\} &\longleftrightarrow \star||\star\\
\{2,2\} &\longleftrightarrow |\,\star\star|\\
\{2,3\} &\longleftrightarrow |\,\star\,|\star\\
\{3,3\} &\longleftrightarrow ||\,\star\star
\end{aligned}
$$

Hence $\left(\!\binom{3}{2}\!\right) = \binom{4}{2} = 6$, as found at the beginning: choose the 2 positions of the stars among 4.

# Application: integer solutions of linear equations

**Question:** how many non-negative integer solutions does the equation

$$x_1 + x_2 + \cdots + x_n = k$$

have?

**Theorem**

The number of non-negative integer solutions of $x_1 + x_2 + \cdots + x_n = k$ is

$$\left(\!\binom{n}{k}\!\right) = \binom{k+n-1}{k}.$$

# Worksheet 1 – Exercise 41

**✎ Exercise (Worksheet 1, Exercise 41 ⋆)**

Give the number of solutions in $\mathbb{N} \cup \{0\}$ for:

1. $x + y + z = 10$ where $x \geq 2$, $y \geq 1$, $z \geq 2$.
2. $x + y + z + t = 10$ where $x \geq 2$, $y \geq 1$, $t \geq 2$.
3. $x_1 + x_2 + \cdots + x_r \leq n$.
4. $x_1 + x_2 + \cdots + x_r \leq n$ where $x_1 \geq 1$, $x_2 \geq 2$, $\ldots$, $x_r \geq r$.

# Worksheet 1 – Exercise 41

**✎ Exercise (Worksheet 1, Exercise 41 ⋆)**

Give the number of solutions in $\mathbb{N} \cup \{0\}$ for:

1. $x + y + z = 10$ where $x \geq 2$, $y \geq 1$, $z \geq 2$.
2. $x + y + z + t = 10$ where $x \geq 2$, $y \geq 1$, $t \geq 2$.
3. $x_1 + x_2 + \cdots + x_r \leq n$.
4. $x_1 + x_2 + \cdots + x_r \leq n$ where $x_1 \geq 1$, $x_2 \geq 2$, …, $x_r \geq r$.

**Two tools**

- A **lower bound** $x \geq a$: put $x' = x - a \geq 0$ (give the minimum first).
- An **inequality** $\leq n$: a **slack variable**, or **cases** on the total $0, \ldots, n$.

# Exercise 41: question 1

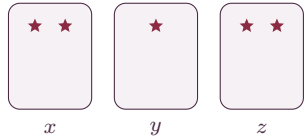
**Solution (1)** $x + y + z = 10, \quad x \geq 2, \; y \geq 1, \; z \geq 2$)

Put $x' = x - 2, \quad y' = y - 1, \quad z' = z - 2.$
Then $x', y', z' \geq 0$ and

$$x' + y' + z' = 10 - (2 + 1 + 2) = 5.$$

Give the minimum first

$2 + 1 + 2 = 5$ given first



# Exercise 41: question 1

**Solution (1)** $x + y + z = 10, \quad x \geq 2, \; y \geq 1, \; z \geq 2$)

Put $x' = x - 2, \quad y' = y - 1, \quad z' = z - 2.$
Then $x', y', z' \geq 0$ and

$$x' + y' + z' = 10 - (2 + 1 + 2) = 5.$$

3 variables, sum 5:

$$\binom{5 + 3 - 1}{3 - 1} = \binom{7}{2} = 21.$$

Give the minimum first

$2 + 1 + 2 = 5$ given first




5 stars left, put freely:
5 stars and 2 bars

# Exercise 41: question 2

**Solution (2)** $x + y + z + t = 10, \quad x \geq 2,\ y \geq 1,\ t \geq 2$)

Same change for $x$, $y$, $t$: $x' = x - 2, \quad y' = y - 1, \quad t' = t - 2$.
**Careful:** $z$ has no condition, but it is still a variable: $z' = z \geq 0$.

$$x' + y' + z' + t' = 10 - (2 + 1 + 2) = 5.$$

# Exercise 41: question 2

**Solution (2)** $x+y+z+t=10,\ \ x\geq 2,\ y\geq 1,\ t\geq 2$)

Same change for $x$, $y$, $t$: $x'=x-2,\ \ y'=y-1,\ \ t'=t-2$.
**Careful:** $z$ has no condition, but it is still a variable: $z'=z\geq 0$.

$$x'+y'+z'+t'=10-(2+1+2)=5.$$

$4$ variables, sum $5$:

$$\binom{5+4-1}{4-1}=\binom{8}{3}=56.$$

# Exercise 41: question 3, method 1

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 1)**

Add a **slack variable** $x_{r+1} = n - (x_1 + \cdots + x_r) \geq 0$. Each solution of the inequality gives exactly one solution of

$$x_1 + x_2 + \cdots + x_r + x_{r+1} = n, \qquad x_i \geq 0,$$

# Exercise 41: question 3, method 1

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 1)**

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and conversely. $r+1$ variables, sum $n$: $\binom{n + (r+1) - 1}{(r+1) - 1} = \binom{n+r}{r}$.

# Exercise 41: question 3, method 1

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 1)**

Add a **slack variable** $x_{r+1} = n - (x_1 + \cdots + x_r) \geq 0$. Each solution of the inequality gives exactly one solution of

$$x_1 + x_2 + \cdots + x_r + x_{r+1} = n, \qquad x_i \geq 0,$$

and conversely. $r+1$ variables, sum $n$: $\binom{n + (r+1) - 1}{(r+1) - 1} = \binom{n+r}{r}$.

Check: $r = 2$, $n = 1$: $(0,0), (1,0), (0,1)$, that is $3 = \binom{3}{2}$.

# Exercise 41: question 3, method 2

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 2)**

The total $x_1 + \cdots + x_r$ takes one of the values $i = 0, 1, \ldots, n$.

# Exercise 41: question 3, method 2

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 2)**

The total $x_1 + \cdots + x_r$ takes one of the values $i = 0, 1, \ldots, n$.

- For a **fixed** value $i$: $x_1 + \cdots + x_r = i$ has $\binom{i+r-1}{r-1}$ solutions (stars and bars).

# Exercise 41: question 3, method 2

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 2)**

The total $x_1 + \cdots + x_r$ takes one of the values $i = 0, 1, \ldots, n$.

- For a **fixed** value $i$: $x_1 + \cdots + x_r = i$ has $\binom{i+r-1}{r-1}$ solutions (stars and bars).
- The cases $i = 0, 1, \ldots, n$ are disjoint, so we **add** them (addition principle):

$$N = \sum_{i=0}^{n} \binom{i+r-1}{r-1} = \binom{r-1}{r-1} + \binom{r}{r-1} + \cdots + \binom{n+r-1}{r-1}.$$

# Exercise 41: question 3, method 2

**Solution (3) $x_1 + x_2 + \cdots + x_r \leq n$ – method 2)**

The total $x_1 + \cdots + x_r$ takes one of the values $i = 0, 1, \ldots, n$.

- For a **fixed** value $i$: $x_1 + \cdots + x_r = i$ has $\binom{i+r-1}{r-1}$ solutions (stars and bars).
- The cases $i = 0, 1, \ldots, n$ are disjoint, so we **add** them (addition principle):

$$N = \sum_{i=0}^{n} \binom{i+r-1}{r-1} = \binom{r-1}{r-1} + \binom{r}{r-1} + \cdots + \binom{n+r-1}{r-1}.$$

Check: $r = 2$, $n = 1$: $\binom{1}{1} + \binom{2}{1} = 1 + 2 = 3$, as with method 1.

# Exercise 41: question 3, method 2 (the sum)

**Solution (The sum telescopes)**

Pascal's rule $\binom{i+r}{r} = \binom{i+r-1}{r-1} + \binom{i+r-1}{r}$ gives $\binom{i+r-1}{r-1} = \binom{i+r}{r} - \binom{i+r-1}{r}$. So

$$N = \left[\binom{r}{r} - \binom{r-1}{r}\right] + \left[\binom{r+1}{r} - \binom{r}{r}\right] + \cdots + \left[\binom{n+r}{r} - \binom{n+r-1}{r}\right]$$

# Exercise 41: question 3, method 2 (the sum)

**Solution (The sum telescopes)**

Pascal's rule $\binom{i+r}{r} = \binom{i+r-1}{r-1} + \binom{i+r-1}{r}$ gives $\binom{i+r-1}{r-1} = \binom{i+r}{r} - \binom{i+r-1}{r}$. So

$$N = \left[\binom{r}{r} - \binom{r-1}{r}\right] + \left[\binom{r+1}{r} - \binom{r}{r}\right] + \cdots + \left[\binom{n+r}{r} - \binom{n+r-1}{r}\right]$$

Everything cancels except the last and the first terms, and $\binom{r-1}{r} = 0$:

$$N = \binom{n+r}{r} - \binom{r-1}{r} = \binom{n+r}{r}.$$

# Exercise 41: question 3, method 2 (the sum)

**Solution (The sum telescopes)**

Pascal's rule $\binom{i+r}{r} = \binom{i+r-1}{r-1} + \binom{i+r-1}{r}$ gives $\binom{i+r-1}{r-1} = \binom{i+r}{r} - \binom{i+r-1}{r}$. So

$$N = \left[\binom{r}{r} - \binom{r-1}{r}\right] + \left[\binom{r+1}{r} - \binom{r}{r}\right] + \cdots + \left[\binom{n+r}{r} - \binom{n+r-1}{r}\right]$$

Everything cancels except the last and the first terms, and $\binom{r-1}{r} = 0$:

$$N = \binom{n+r}{r} - \binom{r-1}{r} = \binom{n+r}{r}.$$

**A free identity:** same solutions, two methods, so $\sum_{i=0}^{n} \binom{i+r-1}{r-1} = \binom{n+r}{r}$ (**hockey-stick identity**).

# Exercise 41: question 4

**Solution (4) the same with $x_1 \geq 1$, $x_2 \geq 2$, $\ldots$, $x_r \geq r$)**

Put $y_i = x_i - i \geq 0$. Since $1 + 2 + \cdots + r = \frac{r(r+1)}{2}$, the inequality becomes

$$y_1 + y_2 + \cdots + y_r \leq n - \frac{r(r+1)}{2} =: m.$$

# Exercise 41: question 4

**Solution (4) the same with $x_1 \geq 1$, $x_2 \geq 2$, $\ldots$, $x_r \geq r$)**

Put $y_i = x_i - i \geq 0$. Since $1 + 2 + \cdots + r = \frac{r(r+1)}{2}$, the inequality becomes

$$y_1 + y_2 + \cdots + y_r \leq n - \frac{r(r+1)}{2} =: m.$$

- If $m < 0$, that is $n < \frac{r(r+1)}{2}$: no solution.
- If $m \geq 0$: by question 3, $\binom{m+r}{r} = \binom{n - \frac{r(r+1)}{2} + r}{r}$.

# Exercise 41: question 4

**Solution (4) the same with $x_1 \geq 1$, $x_2 \geq 2$, $\ldots$, $x_r \geq r$)**

Put $y_i = x_i - i \geq 0$. Since $1 + 2 + \cdots + r = \frac{r(r+1)}{2}$, the inequality becomes

$$y_1 + y_2 + \cdots + y_r \leq n - \frac{r(r+1)}{2} =: m.$$

- If $m < 0$, that is $n < \frac{r(r+1)}{2}$: no solution.
- If $m \geq 0$: by question 3, $\binom{m+r}{r} = \binom{n - \frac{r(r+1)}{2} + r}{r}$.

Check: $r = 2$, $n = 4$: $m = 4 - 3 = 1$, $\binom{3}{2} = 3$: $(1,2), (1,3), (2,2)$.

# Worksheet 1 – Exercise 44

✎ **Exercise (Worksheet 1, Exercise 44 ⋆)**

Find the number of integer solutions to the equation $x_1+x_2+x_3+x_4+x_5+x_6 = 60$, in each of the following cases:

(i) $x_i \geq i-1$ for each $i = 1, 2, \ldots, 6$;

(ii) $x_1 \geq 2$, $x_2 \geq 5$, $2 \leq x_3 \leq 7$, $x_4 \geq 1$, $x_5 \geq 3$ and $x_6 \geq 2$.

# Worksheet 1 – Exercise 44

**✎ Exercise (Worksheet 1, Exercise 44 ⋆)**

Find the number of integer solutions to the equation $x_1+x_2+x_3+x_4+x_5+x_6 = 60$, in each of the following cases:

(i) $x_i \geq i-1$ for each $i = 1, 2, \ldots, 6$;

(ii) $x_1 \geq 2$, $x_2 \geq 5$, $2 \leq x_3 \leq 7$, $x_4 \geq 1$, $x_5 \geq 3$ and $x_6 \geq 2$.

**Idea**

(i) Lower bounds only: change of variables. (ii) There is also an **upper bound** $x_3 \leq 7$: count **all** the solutions, then remove the **bad** ones ($x_3 \geq 8$).

# Exercise 44: part (i)

**Solution ((i) $x_i \geq i-1$)**

Put $y_i = x_i - (i-1) \geq 0$. We subtract $0+1+2+3+4+5 = 15$:

$$y_1 + \cdots + y_6 = 45 \qquad \Longrightarrow \qquad \binom{45+6-1}{6-1} = \binom{50}{5} = 2\,118\,760.$$

# Exercise 44: part (ii)

**Solution ((ii) with $2 \leq x_3 \leq 7$)**

Put $y_1 = x_1 - 2$, $y_2 = x_2 - 5$, $y_3 = x_3 - 2$, $y_4 = x_4 - 1$, $y_5 = x_5 - 3$, $y_6 = x_6 - 2$.
We subtract $2 + 5 + 2 + 1 + 3 + 2 = 15$:

$$y_1 + \cdots + y_6 = 45, \qquad y_i \geq 0, \qquad y_3 \leq 5.$$

# Exercise 44: part (ii)

**Solution ((ii) with $2 \leq x_3 \leq 7$)**

Put $y_1 = x_1 - 2$, $y_2 = x_2 - 5$, $y_3 = x_3 - 2$, $y_4 = x_4 - 1$, $y_5 = x_5 - 3$, $y_6 = x_6 - 2$.
We subtract $2 + 5 + 2 + 1 + 3 + 2 = 15$:

$$y_1 + \cdots + y_6 = 45, \qquad y_i \geq 0, \qquad y_3 \leq 5.$$

- **All** solutions (forget $y_3 \leq 5$): $\binom{50}{5} = 2\,118\,760$.

# Exercise 44: part (ii)

**Solution ((ii) with $2 \le x_3 \le 7$)**

Put $y_1 = x_1 - 2$, $y_2 = x_2 - 5$, $y_3 = x_3 - 2$, $y_4 = x_4 - 1$, $y_5 = x_5 - 3$, $y_6 = x_6 - 2$.
We subtract $2 + 5 + 2 + 1 + 3 + 2 = 15$:

$$y_1 + \cdots + y_6 = 45, \qquad y_i \ge 0, \qquad y_3 \le 5.$$

- **All** solutions (forget $y_3 \le 5$): $\binom{50}{5} = 2\,118\,760$.
- **Bad** solutions ($y_3 \ge 6$): put $z_3 = y_3 - 6 \ge 0$, the sum becomes $39$: $\binom{39+5}{5} = \binom{44}{5} = 1\,086\,008$.

# Exercise 44: part (ii)

**Solution ((ii) with $2 \le x_3 \le 7$)**

Put $y_1 = x_1 - 2$, $y_2 = x_2 - 5$, $y_3 = x_3 - 2$, $y_4 = x_4 - 1$, $y_5 = x_5 - 3$, $y_6 = x_6 - 2$.
We subtract $2 + 5 + 2 + 1 + 3 + 2 = 15$:

$$y_1 + \cdots + y_6 = 45, \qquad y_i \ge 0, \qquad y_3 \le 5.$$

- **All** solutions (forget $y_3 \le 5$): $\binom{50}{5} = 2\,118\,760$.
- **Bad** solutions ($y_3 \ge 6$): put $z_3 = y_3 - 6 \ge 0$, the sum becomes $39$: $\binom{39+5}{5} = \binom{44}{5} = 1\,086\,008$.
- **Good** solutions: $\binom{50}{5} - \binom{44}{5} = 2\,118\,760 - 1\,086\,008 = 1\,032\,752$.

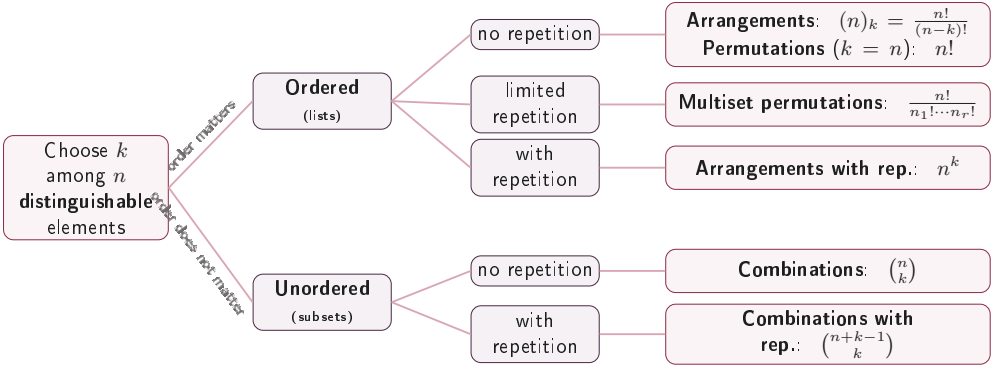
04

# Summary

Four ways to select $k$ elements from $n$

# The tree of notions: summary

All the leaves are filled: the whole chapter in one picture.



# Selecting $k$ elements from a set of $n$ elements

| | Without repetition | With repetition |
|---|---|---|
| **Ordered** | $(n)_k = \dfrac{n!}{(n-k)!}$ | $n^k$ |
| **Unordered** | $\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$ | $\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}$ |

Plus: permutations $n!$ and multinomial coefficients $\dfrac{n!}{n_1! \cdots n_k!}$ for multisets.

# Back to the opening questions

- Arrangements of "NHSM": $4! = 24$.
- Gold, silver, bronze among 12 students: $(12)_3 = 1320$.
- 6 balls into 4 boxes: $4^6 = 4096$ if the balls are distinct, but $\left(\!\binom{4}{6}\!\right) = \binom{9}{3} = 84$ if they are identical. Distinguishability matters!
- 5 distinct colors on 5 chairs in a row: $5! = 120$.
- Passwords of length $n$: $26^n$.
- 7 tasks, 3 employees, each gets at least one task: $1806$. This needs the **Inclusion–Exclusion Principle** (next chapter).

# Thank you!

Questions?

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